MetidaNCA validation report

Vladimir Arnautov

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```
acceptance_verdict (generic function with 1 method)
```

1 Introduction and package description

This is Non-compartment anlysis software. The package is designed for batch processing of pharmacokinetic data.

See documentation:

- Dev: https://pharmcat.github.io/MetidaNCA.jl/dev/
- Stable: https://pharmcat.github.io/MetidaNCA.jl/stable/

1.1 Validation purpose

The main validation purpose is confirmation by examination and provision of objective evidence that software specifications conform to user needs and intended uses, and that the particular requirements implemented through software can be consistently fulfilled.

1.2 Acceptance criteria

1.2.1 Kinds of evidence

Comparisons in this report rest on four kinds of reference. The kinds are not interchangeable. The chapter *Validation coverage* reports, for every parameter and every capability, which kind applies.

- **External.** The corresponding value from an independent implementation, Phoenix WinNonlin 8.0 (see Appendix 2). Most comparisons of the report are of this kind.
- **Analytic.** A closed-form value. The reference is exact and belongs to no implementation. The chapter on the integration rules rests on references of this kind.
- **Derived.** The value computed from the identity that defines the parameter, where the parameters on the right-hand side are themselves qualified. A derived comparison inherits whatever its inputs inherit and establishes nothing about the inputs themselves. It is used where no other program reports the parameter.
- **Internal.** A value this package produced, carried forward unchanged. An internal baseline establishes that behaviour has not changed. An internal baseline is not evidence of correctness. It is used only where no external reference can be obtained, which in this report means the `:luldt` integration rule, for which Phoenix implements no counterpart. Wherever an internal baseline is used, the report states the fact at the point of use.

Graphics are qualified operationally. The figure is reproduced for visual verification, and the data the figure carries is compared with the data the package computed.

1.2.2 The criterion

A single comparison is accepted when

$$|x_{MetidaNCA} - x_{ref}| \leq r_{tol} \times |x_{ref}| + a_{tol}$$

with $r_{tol} = 0.0001$ and $a_{tol} = 0.0$, that is, agreement to four significant digits. The same criterion is applied to every kind of reference.

External reference values are transcribed with six significant digits. The attainable agreement is therefore bounded by the transcription of the reference and not by the computation. The largest deviation observed over the external scenarios of this report is of the order of 10^{-7}, which leaves about three orders of magnitude of margin to the criterion.

Against a closed form the bound is machine precision. The deviations observed are of the order of 10^{-16}, that is, twelve orders of magnitude below the criterion. The agreement attained for each comparison is reported in the tables of the integration-rules chapter.

The checks on figures require exact equality. A figure carries the computed numbers without rounding them.

The report is accepted only when **all** of the following hold:

- every comparison is within the criteria above.
- the package test suite passes (see Testing results).
- every required scenario reported its results.

The third condition covers a failure the first two cannot detect. The document generator renders a failing code chunk into the document and carries on, which drops a results table without any other sign. The same applies to the Conclusion, which marks itself once it has rendered.

1.2.3 Where the verdict is

This report is produced whatever the outcome of the comparisons, and it carries its own verdict. Each comparison is recorded, as it is made, in `docs/src/validation_acceptance.tsv`. The Conclusion section states the verdict, summarises every scenario and lists **every divergence** with its parameter, subject, computed value, reference value, difference, relative error and the criterion it was judged against. The same journal is written next to this report as `docs/src/validation_divergences.tsv`, so it can be read without opening the PDF. The file is written even when nothing diverged, so an empty journal states that no divergence was recorded.

Independently of the report, `acceptance_verdict()` in `docs/acceptance.jl` is called outside the document generator, after the report is built, and aborts the documentation build on a negative verdict. The document generator swallows errors raised inside a code chunk, so a verdict computed only inside the report could never fail. Running the build with `VALIDATION_STRICT=false` suppresses the abort and leaves the report for inspection. The verdict and the journal are unaffected by that setting.

1.3 Requirements

- Julia 1.8 (or higher) installed for Operating System/OS Version/Architecture in Tier 1 list

Tier 1: Julia is guaranteed to build from source and pass all tests on these platforms when built with the default options. Official binaries are always available and CI is run on every commit to ensure support is actively maintained.

1.4 Developer software life cycle

- Development stage
- Testing procedures development
- Performing testing procedures on local machine
- Push to development branch
- Make pull request to main branch
- Performing testing procedures with GitHub Actions
- Make pull request to the official registry of general Julia packages (if nessesary)
- Make release (if previous completed)

1.4.1 Versions

- X.Y.Z - patch release (no breaking changes)
- X.Y.0 - minor release (may include breaking changes if X = 0)

- X.0.0 - major release (breaking changes, changes in public API)
- 0.#.# - no stable public API
- 1.#.# or higher - stable public API

1.5 Build support

1.5.1 Tier 1

- julia-version: 1.8, 1
- julia-arch: x64
- os: ubuntu-latest, macOS-latest, windows-latest

2 Installation

2.1 System information

- Julia version: v"1.13.2"
- Current machine: "x86_64-linux-gnu"

2.2 Installation method

MetidaNCA.jl can be installed by executing the following command in REPL:

```
import Pkg; Pkg.add("MetidaNCA")
```

2.3 Version check

The installation process is checking within each testing job via GitHub Actions. Also GitHub Action chek performed before merging into JuliaRegistries/General repository (see Automatic merging of pull requests).

Current package version: Error: UndefVarError: `ver` not defined in `Main.var"##WeaveSandBox#145"` Suggestion: add an appropriate import or assignment. This global was declared but not assigned.

2.3.1 Dependencies

The table lists every dependency of MetidaNCA with the version loaded during this build. A direct dependency is declared in the `[deps]` section of the package `Project.toml`, and the bound declared for it in the `[compat]` section is given in the same row. An indirect dependency is loaded through a direct one and carries no bound of its own. The resolver installs a version that satisfies every bound, and a version outside the bounds leaves the environment unresolvable, so the build does not start.

Packages used by this report and by the test suite are not dependencies of the package and are not listed. The bound on the Julia version is Error: UndefVarError: `JULIA_BOUND` not defined in `Main.var"##WeaveSandBox#145"` Suggestion: check for spelling errors or missing imports., and the version in use is stated under System information.

Dependencies loaded during this build: Error: UndefVarError: `dfdep` not defined in `Main.var"##WeaveSandBox#145"` Suggestion: check for spelling errors or missing imports., of which direct: Error: UndefVarError: `dfdep` not defined in `Main.var"##WeaveSandBox#145"` Suggestion: check for spelling errors or missing imports.. The list includes the standard libraries of Julia that the dependencies load, and their versions are the versions of the Julia release in use.

Error: UndefVarError: `dfdep` not defined in `Main.var"##WeaveSandBox#145"` Suggestion: check for spelling errors or missing imports.

Table: Dependencies of MetidaNCA and the versions loaded during this build

3 Operation qualification

This part of validation based on testing procedures entails running software products under known conditions with defined inputs and documented outcomes that can be compared to their predefined expectations. All documented public API included in testing procedures and part of critical internal methods. Testing procedures can be found in `test` directory.

3.1 Coverage

Code coverage report available on Codecov.io. Test procedures include all public API methods check.

- Coverage goal: >= 90.0%

3.2 Data

Validation data available in the repository and included in the package. See Appendix 1.

3.3 Testing results

The outcome is recorded and enforced: a failing test suite makes `acceptance_verdict()` abort the build (see Acceptance criteria). The `try/catch` below exists only because the document generator does not propagate errors raised inside a chunk.

```
test_passed = try
    Pkg.test("MetidaNCA")
    true
catch err
    @error "MetidaNCA test suite failed" exception = (err, catch_backtrace())
    false
end
acceptance_mark("PKG-TEST", test_passed, "Pkg.test(MetidaNCA)")
```

```
true
```

4 Performance qualification

Purpose of this testing procedures to demonstrate performance for some critical tasks. Results from MetidaNCA compared with Phoenix WinNonlin 8.0 results, see Appendix 2.

4.1 Parameter's names description

Table 1: Parameter description

Name	Description
Cmax	Maximum concentration
Tmax	Time at Cmax
Cdose	Concentration at dose time
Clast	Last non-zero concentration
AUClast	AUC to Clast
AUMClast	AUMC to Clast
AUCall	AUC with all values
Rsq	r square
ARsq	Adjusted r square
Kel	Terminal elimination constant
HL	Half live or T1/2
LZint	Intercept
Clast_pred	Predicted Clast
AUCinf	AUC extrapolated to infinity
AUCpct	Percentage AUClast from AUCinf
MRTlast	Mean Residence Time (last)
MRTinf	Mean residence time (MRT) extrapolated to infinity
MRTinf_pred	Mean residence time (inf) predicted
Clinf	Clearence
Vzinf	Volume of distribution
Ctaumin	Minimum concentration between dose time and dose time + Tau
Cavg	AUCtau divided by Tau.
Accind	Accumulation Index
Fluc	Fluctuation
AUCtau	AUC in Tau range
AUMCtau	AUMC in Tau range
MRTtauinf	MRT based on Tau
MRTtauinf_pred	Mean residence time (inf) predicted based on Tau
Cltau	Clearence in Tau range
Vztau	Volume of distribution in Tau range

Note!!! When `tau` specified additional MRT parameters calculated: `MRTtauinf`, `MRTtauinf_pred`; this
parameters are equal with Pjoenix WinNonlin parameters: `MRTINF_obs` and `MRTINF_pred` calculated for steady-state data. Parameters `MRTinf` and `MRTinf_pred` calculated as usual.

4.2 Output example

Import data:

```
pkdata2  = CSV.File(
joinpath(dirname(pathof(MetidaNCA)), "..", "test", "csv", "pkdata2.csv")
) |> DataFrame
ds = pkimport(pkdata2, :Time, :Concentration, [:Subject, :Formulation];
dosetime = DoseTime(dose = 100, time = 0))
sort!(ds, :Subject)
```

Execute NCA:

```
MetidaNCA.nca!(ds[1], adm = :ev, calcm = :lint, verbose = 1)
```

```
  Non-compartmental Pharmacokinetic Analysis
    Subject: Subject => 1; Formulation => T;
    Settings:
    Method: lint; Dose: 100; Dose time: 0
------ ------- ------- -------- ---------- ---------- ------
 Time   Conc.     AUC      AUC        AUMC        AUMC   Info
                         (cum.)                 (cum.)
------ ------- ------- -------- ---------- ---------- ------
    0       0       0        0          0          0      D
  0.5   178.9   44.74    44.74      22.37      22.37
    1   190.9   92.45    137.2      70.09      92.45
  1.5   164.9   88.95    226.1      109.6        202
    2     140   76.22    302.4      131.8      333.8
  2.5   129.6   67.39    369.8        151      484.8
    3   131.4   65.24      435      179.5      664.3
    4   150.9   141.1    576.1      498.8       1163
    5   121.2     136    712.1      604.8       1768
    6   139.2   130.2    842.4      720.8       2489
    8   128.5   267.7     1110       1864       4352
   10   143.2   271.8     1382       2461       6813      E
   12     145   288.2     1670       3172       9985      E
   24   133.2    1669     3339   2.961e+04   3.96e+04      E
   48   137.3    3245     6584   1.174e+05   1.57e+05      E
   72   112.8    3001     9585   1.766e+05   3.336e+05     E
------ ------- ------- -------- ---------- ---------- ------

    Cdose: 0.0, Dose time: 0
    Kel start: 10.0; end: 72.0

 PK/PD subject NCA result
------------- -------------
    Parameter         Value
------------- -------------
         Rsqn           3.0
       Vzlast       3.08222
         Tmax           1.0
```

```
 AUMCinf_pred     1.30681e7
  MRTinf_pred       295.374
         ARsq      0.714769
      AUClast       9585.42
       MRTinf       293.162
  AUCinf_pred       44242.6
           LZ   -0.00338474
        LZint       5.00849
       Obsnum          16.0
        Clast       112.846
         Dose         100.0
         Tlag           0.0
        Cdose           0.0

-------------- -------------
              17 rows omitted
```

4.3 Results

4.3.1 Linear-trapezoidal rule; Extravascular; Dosetime 0.0; No Tau; Dose 100

Code:

```
nca!(ds, adm = :ev, calcm = :lint)
```

Table 2: Plasma data results, Linear-trapezoidal rule, Extravascular

Parameter	Subject	Value	Reference	Difference	Status
ARsq	1	0.714769	0.714769	-3.89242e-9	PASS
ARsq	2	0.990351	0.990351	4.11874e-9	PASS
ARsq	3	0.776307	0.776307	-4.40849e-9	PASS
ARsq	4	0.837717	0.837717	6.63705e-11	PASS
ARsq	5	0.82892	0.82892	3.67764e-9	PASS
ARsq	6	0.925179	0.925179	-3.46941e-10	PASS
ARsq	7	0.960416	0.960416	-4.84619e-10	PASS
ARsq	8	0.921954	0.921954	-2.59589e-10	PASS
ARsq	9	0.921307	0.921307	2.29085e-9	PASS
ARsq	10	0.863912	0.863912	-4.57725e-9	PASS
AUCall	1	9585.42	9585.42	-5.0e-5	PASS
AUCall	2	10112.2	10112.2	-0.0005	PASS
AUCall	3	5396.55	5396.55	-5.0e-5	PASS
AUCall	4	9317.84	9317.84	-5.0e-5	PASS
AUCall	5	9561.26	9561.26	0.0	PASS
AUCall	6	6966.6	6966.6	9.09495e-13	PASS
AUCall	7	7029.57	7029.57	0.0	PASS
AUCall	8	7110.67	7110.67	-9.09495e-13	PASS
AUCall	9	8315.08	8315.08	-5.0e-5	PASS
AUCall	10	5620.89	5620.89	0.0	PASS
AUCinf	1	42925.0	42925.0	7.26088e-5	PASS
AUCinf	2	16154.9	16154.9	0.000134568	PASS
AUCinf	3	26026.2	26026.2	-0.000441279	PASS
AUCinf	4	22004.1	22004.1	-0.000131232	PASS
AUCinf	5	25820.3	25820.3	-0.000104764	PASS
AUCinf	6	16001.8	16001.8	-0.000256431	PASS
AUCinf	7	11689.0	11689.0	-0.000326636	PASS
AUCinf	8	15446.2	15446.2	0.000305556	PASS
AUCinf	9	24865.2	24865.2	-3.66208e-5	PASS
AUCinf	10	8075.32	8075.32	-3.5589e-5	PASS
AUCinf_pred	1	44242.6	44242.6	0.00026702	PASS
AUCinf_pred	2	15963.2	15963.2	-5.36657e-5	PASS
AUCinf_pred	3	25731.5	25731.5	0.000158813	PASS
AUCinf_pred	4	22412.5	22412.5	4.14188e-7	PASS
AUCinf_pred	5	25016.0	25016.0	2.34391e-5	PASS
AUCinf_pred	6	16318.8	16318.8	0.000318763	PASS
AUCinf_pred	7	11939.5	11939.5	-0.000426398	PASS
AUCinf_pred	8	15576.9	15576.9	0.00020719	PASS
AUCinf_pred	9	24922.2	24922.2	-0.000375061	PASS
AUCinf_pred	10	7878.69	7878.69	1.0255e-5	PASS
AUClast	1	9585.42	9585.42	-5.0e-5	PASS
AUClast	2	10112.2	10112.2	-0.0005	PASS
AUClast	3	5396.55	5396.55	-5.0e-5	PASS
AUClast	4	9317.84	9317.84	-5.0e-5	PASS
AUClast	5	9561.26	9561.26	0.0	PASS
AUClast	6	6966.6	6966.6	9.09495e-13	PASS
AUClast	7	7029.57	7029.57	0.0	PASS

Parameter	Subject	Value	Reference	Difference	Status
AUClast	8	7110.67	7110.67	-9.09495e-13	PASS
AUClast	9	8315.08	8315.08	-5.0e-5	PASS
AUClast	10	5620.89	5620.89	0.0	PASS
AUCpct	1	77.6694	77.6694	-3.34814e-7	PASS
AUCpct	2	37.405	37.405	-3.2005e-7	PASS
AUCpct	3	79.2649	79.2649	9.4124e-8	PASS
AUCpct	4	57.6541	57.654	2.82991e-7	PASS
AUCpct	5	62.97	62.97	-2.74068e-7	PASS
AUCpct	6	56.4636	56.4636	-1.88346e-7	PASS
AUCpct	7	39.8614	39.8614	3.80099e-7	PASS
AUCpct	8	53.9649	53.9649	-7.63594e-9	PASS
AUCpct	9	66.5594	66.5594	-7.75843e-8	PASS
AUCpct	10	30.3942	30.3942	-3.49818e-7	PASS
AUMCinf	1	1.2584e7	1.2584e7	-0.063378	PASS
AUMCinf	2	1.16215e6	1.16215e6	0.0262866	PASS
AUMCinf	3	7.93905e6	7.93905e6	-0.0331478	PASS
AUMCinf	4	2.87593e6	2.87593e6	0.0450708	PASS
AUMCinf	5	3.87219e6	3.87218e6	0.0136558	PASS
AUMCinf	6	2.05208e6	2.05208e6	-0.0403706	PASS
AUMCinf	7	929251.0	929251.0	-0.00247654	PASS
AUMCinf	8	1.77411e6	1.77411e6	-0.0491652	PASS
AUMCinf	9	4.4006e6	4.4006e6	-0.0252583	PASS
AUMCinf	10	474397.0	474397.0	0.00441634	PASS
AUMCinf_pred	1	1.30681e7	1.30681e7	-0.259125	PASS
AUMCinf_pred	2	1.13476e6	1.13476e6	-0.0449258	PASS
AUMCinf_pred	3	7.8283e6	7.8283e6	-0.0390742	PASS
AUMCinf_pred	4	2.95841e6	2.95841e6	-0.0391127	PASS
AUMCinf_pred	5	3.69624e6	3.69624e6	-0.0278232	PASS
AUMCinf_pred	6	2.11613e6	2.11613e6	0.0466328	PASS
AUMCinf_pred	7	967402.0	967402.0	0.00443394	PASS
AUMCinf_pred	8	1.79816e6	1.79816e6	-0.0480829	PASS
AUMCinf_pred	9	4.4148e6	4.4148e6	0.0328207	PASS
AUMCinf_pred	10	448799.0	448799.0	-0.0021186	PASS
AUMClast	1	333582.0	333582.0	0.00075	PASS
AUMClast	2	298701.0	298701.0	-0.0015	PASS
AUMClast	3	186032.0	186032.0	-0.00475	PASS
AUMClast	4	313956.0	313956.0	0.00475	PASS
AUMClast	5	315182.0	315182.0	0.0025	PASS
AUMClast	6	226977.0	226977.0	0.00075	PASS
AUMClast	7	219798.0	219798.0	-0.00275	PASS
AUMClast	8	240526.0	240526.0	0.00375	PASS
AUMClast	9	277614.0	277614.0	-0.00225	PASS
AUMClast	10	154893.0	154893.0	0.0005	PASS
Clast	1	112.846	112.846	0.0	PASS
Clast	2	85.241	85.241	0.0	PASS
Clast	3	67.901	67.901	0.0	PASS
Clast	4	97.625	97.625	0.0	PASS
Clast	5	110.778	110.778	0.0	PASS
Clast	6	69.501	69.501	0.0	PASS
Clast	7	58.051	58.051	0.0	PASS
Clast	8	74.437	74.437	0.0	PASS
Clast	9	93.44	93.44	0.0	PASS
Clast	10	42.191	42.191	0.0	PASS
Clast_pred	1	117.306	117.306	-1.09686e-7	PASS
Clast_pred	2	82.5367	82.5367	-1.86741e-7	PASS
Clast_pred	3	66.9311	66.9311	-5.84471e-8	PASS
Clast_pred	4	100.768	100.768	3.50061e-6	PASS
Clast_pred	5	105.298	105.298	5.80678e-7	PASS

Parameter	Subject	Value	Reference	Difference	Status
Clast_pred	6	71.9399	71.9399	9.56122e-9	PASS
Clast_pred	7	61.1727	61.1727	3.07644e-7	PASS
Clast_pred	8	75.6043	75.6043	-3.56254e-7	PASS
Clast_pred	9	93.7618	93.7618	-4.23203e-7	PASS
Clast_pred	10	38.8109	38.8109	3.50002e-7	PASS
Clinf	1	0.00232964	0.00232964	-5.93786e-12	PASS
Clinf	2	0.00619006	0.00619006	1.53062e-11	PASS
Clinf	3	0.00384228	0.00384228	-1.69892e-11	PASS
Clinf	4	0.00454461	0.00454461	-3.32735e-11	PASS
Clinf	5	0.00387293	0.00387293	-4.10053e-11	PASS
Clinf	6	0.00624931	0.00624931	-2.42369e-11	PASS
Clinf	7	0.00855509	0.00855509	-3.94753e-12	PASS
Clinf	8	0.00647408	0.00647408	2.13402e-11	PASS
Clinf	9	0.00402168	0.00402168	-8.96384e-12	PASS
Clinf	10	0.0123834	0.0123834	-1.9304e-10	PASS
Cmax	1	190.869	190.869	0.0	PASS
Cmax	2	261.177	261.177	0.0	PASS
Cmax	3	105.345	105.345	0.0	PASS
Cmax	4	208.542	208.542	0.0	PASS
Cmax	5	169.334	169.334	0.0	PASS
Cmax	6	154.648	154.648	0.0	PASS
Cmax	7	153.254	153.254	0.0	PASS
Cmax	8	138.327	138.327	0.0	PASS
Cmax	9	167.347	167.347	0.0	PASS
Cmax	10	125.482	125.482	0.0	PASS
Dose	1	100.0	100.0	0.0	PASS
Dose	2	100.0	100.0	0.0	PASS
Dose	3	100.0	100.0	0.0	PASS
Dose	4	100.0	100.0	0.0	PASS
Dose	5	100.0	100.0	0.0	PASS
Dose	6	100.0	100.0	0.0	PASS
Dose	7	100.0	100.0	0.0	PASS
Dose	8	100.0	100.0	0.0	PASS
Dose	9	100.0	100.0	0.0	PASS
Dose	10	100.0	100.0	0.0	PASS
HL	1	204.786	204.786	-3.0616e-6	PASS
HL	2	49.1374	49.1374	4.37811e-7	PASS
HL	3	210.591	210.591	-3.91935e-6	PASS
HL	4	90.0736	90.0736	1.94606e-8	PASS
HL	5	101.734	101.734	1.56651e-6	PASS
HL	6	90.1095	90.1094	5.7666e-8	PASS
HL	7	55.6345	55.6345	3.82012e-7	PASS
HL	8	77.6194	77.6194	3.08325e-7	PASS
HL	9	122.771	122.771	-5.00549e-7	PASS
HL	10	40.3233	40.3233	4.40951e-7	PASS
Kel	1	0.00338474	0.00338474	4.00078e-11	PASS
Kel	2	0.0141063	0.0141063	-5.67502e-11	PASS
Kel	3	0.00329143	0.00329143	-2.75071e-11	PASS
Kel	4	0.00769534	0.00769534	2.29811e-11	PASS
Kel	5	0.00681333	0.00681333	1.1549e-11	PASS
Kel	6	0.00769228	0.00769228	-3.33622e-11	PASS
Kel	7	0.012459	0.012459	-2.32353e-11	PASS
Kel	8	0.00893008	0.00893008	9.67252e-12	PASS
Kel	9	0.00564586	0.00564586	1.87097e-11	PASS
Kel	10	0.0171897	0.0171897	-1.6958e-10	PASS
LZint	1	5.00849	5.00849	-7.44672e-9	PASS
LZint	2	5.4289	5.4289	-4.59704e-9	PASS
LZint	3	4.44065	4.44065	-2.44468e-8	PASS

Parameter	Subject	Value	Reference	Difference	Status
LZint	4	5.16688	5.16688	-3.09526e-8	PASS
LZint	5	5.14736	5.14736	-2.02572e-8	PASS
LZint	6	4.82968	4.82968	4.01706e-8	PASS
LZint	7	5.01075	5.01075	-2.03852e-8	PASS
LZint	8	4.96848	4.96848	-2.75635e-9	PASS
LZint	9	4.94726	4.94726	-1.20523e-8	PASS
LZint	10	4.89636	4.89636	-1.2117e-8	PASS
MRTinf	1	293.162	293.162	4.50211e-6	PASS
MRTinf	2	71.9379	71.9379	1.93456e-7	PASS
MRTinf	3	305.041	305.041	1.53794e-6	PASS
MRTinf	4	130.7	130.7	4.95035e-6	PASS
MRTinf	5	149.967	149.967	-7.86749e-7	PASS
MRTinf	6	128.241	128.241	-3.24289e-6	PASS
MRTinf	7	79.4983	79.4983	1.9508e-7	PASS
MRTinf	8	114.857	114.857	1.8871e-6	PASS
MRTinf	9	176.978	176.978	-2.43482e-6	PASS
MRTinf	10	58.7464	58.7464	-6.69597e-8	PASS
MRTinf_pred	1	295.374	295.374	-3.07788e-6	PASS
MRTinf_pred	2	71.0858	71.0858	2.67045e-7	PASS
MRTinf_pred	3	304.23	304.23	-2.75788e-6	PASS
MRTinf_pred	4	131.998	131.998	-3.15878e-6	PASS
MRTinf_pred	5	147.755	147.755	-2.88468e-6	PASS
MRTinf_pred	6	129.674	129.674	-1.06251e-7	PASS
MRTinf_pred	7	81.0253	81.0253	1.87663e-7	PASS
MRTinf_pred	8	115.437	115.437	1.9972e-6	PASS
MRTinf_pred	9	177.143	177.143	-9.39318e-7	PASS
MRTinf_pred	10	56.9637	56.9637	-7.27456e-8	PASS
MRTlast	1	34.801	34.801	-2.12453e-7	PASS
MRTlast	2	29.5388	29.5388	4.06497e-8	PASS
MRTlast	3	34.4724	34.4724	2.35113e-7	PASS
MRTlast	4	33.6941	33.6941	1.67704e-7	PASS
MRTlast	5	32.9644	32.9644	2.94104e-9	PASS
MRTlast	6	32.5808	32.5808	4.73046e-7	PASS
MRTlast	7	31.2676	31.2676	-3.34258e-7	PASS
MRTlast	8	33.8261	33.8261	1.75406e-7	PASS
MRTlast	9	33.3868	33.3868	-2.10011e-7	PASS
MRTlast	10	27.5567	27.5567	-2.25887e-7	PASS
NpLZ	1	5.0	5.0	0.0	PASS
NpLZ	2	5.0	5.0	0.0	PASS
NpLZ	3	7.0	7.0	0.0	PASS
NpLZ	4	3.0	3.0	0.0	PASS
NpLZ	5	8.0	8.0	0.0	PASS
NpLZ	6	4.0	4.0	0.0	PASS
NpLZ	7	5.0	5.0	0.0	PASS
NpLZ	8	4.0	4.0	0.0	PASS
NpLZ	9	4.0	4.0	0.0	PASS
NpLZ	10	9.0	9.0	0.0	PASS
Obsnum	1	16.0	16.0	0.0	PASS
Obsnum	2	16.0	16.0	0.0	PASS
Obsnum	3	16.0	16.0	0.0	PASS
Obsnum	4	16.0	16.0	0.0	PASS
Obsnum	5	16.0	16.0	0.0	PASS
Obsnum	6	16.0	16.0	0.0	PASS
Obsnum	7	16.0	16.0	0.0	PASS
Obsnum	8	16.0	16.0	0.0	PASS
Obsnum	9	16.0	16.0	0.0	PASS
Obsnum	10	16.0	16.0	0.0	PASS
Rsq	1	0.786077	0.786077	-2.91932e-9	PASS

Parameter	Subject	Value	Reference	Difference	Status
Rsq	2	0.992764	0.992764	5.89052e-10	PASS
Rsq	3	0.813589	0.813589	-3.40409e-10	PASS
Rsq	4	0.918859	0.918859	-4.96681e-9	PASS
Rsq	5	0.85336	0.85336	1.72369e-9	PASS
Rsq	6	0.950119	0.950119	-2.31294e-10	PASS
Rsq	7	0.970312	0.970312	4.63654e-9	PASS
Rsq	8	0.947969	0.947969	-1.73059e-10	PASS
Rsq	9	0.947538	0.947538	4.86057e-9	PASS
Rsq	10	0.880923	0.880923	-2.55092e-10	PASS
Tlag	1	0.0	0.0	0.0	PASS
Tlag	2	0.0	0.0	0.0	PASS
Tlag	3	0.0	0.0	0.0	PASS
Tlag	4	0.0	0.0	0.0	PASS
Tlag	5	0.5	0.5	0.0	PASS
Tlag	6	0.0	0.0	0.0	PASS
Tlag	7	0.0	0.0	0.0	PASS
Tlag	8	0.0	0.0	0.0	PASS
Tlag	9	0.0	0.0	0.0	PASS
Tlag	10	0.0	0.0	0.0	PASS
Tlast	1	72.0	72.0	0.0	PASS
Tlast	2	72.0	72.0	0.0	PASS
Tlast	3	72.0	72.0	0.0	PASS
Tlast	4	72.0	72.0	0.0	PASS
Tlast	5	72.0	72.0	0.0	PASS
Tlast	6	72.0	72.0	0.0	PASS
Tlast	7	72.0	72.0	0.0	PASS
Tlast	8	72.0	72.0	0.0	PASS
Tlast	9	72.0	72.0	0.0	PASS
Tlast	10	72.0	72.0	0.0	PASS
Tmax	1	1.0	1.0	0.0	PASS
Tmax	2	1.0	1.0	0.0	PASS
Tmax	3	1.5	1.5	0.0	PASS
Tmax	4	1.0	1.0	0.0	PASS
Tmax	5	4.0	4.0	0.0	PASS
Tmax	6	2.5	2.5	0.0	PASS
Tmax	7	2.5	2.5	0.0	PASS
Tmax	8	4.0	4.0	0.0	PASS
Tmax	9	3.0	3.0	0.0	PASS
Tmax	10	2.0	2.0	0.0	PASS
Vzinf	1	0.688278	0.688278	-3.65182e-9	PASS
Vzinf	2	0.438815	0.438815	4.06237e-9	PASS
Vzinf	3	1.16736	1.16736	2.86303e-8	PASS
Vzinf	4	0.590566	0.590566	-3.8362e-9	PASS
Vzinf	5	0.568434	0.568434	-1.11744e-9	PASS
Vzinf	6	0.812414	0.812414	2.13169e-9	PASS
Vzinf	7	0.686662	0.686662	-8.2846e-12	PASS
Vzinf	8	0.724974	0.724974	4.95686e-9	PASS
Vzinf	9	0.712323	0.712323	-4.58031e-9	PASS
Vzinf	10	0.720395	0.720395	4.47706e-9	PASS

Comparisons: 270. Outside acceptance criteria: 0. Result: PASS.

4.3.2 Linear-Up Log-Down; Extravascular; Dosetime 0.25; Tau 9; Dose 100

Code:

```
setdosetime!(ds, DoseTime(dose = 100, time = 0.25, tau = 9))
nca!(ds, adm = :ev, calcm = :luld)
```

Table 3: Plasma data results, Linear-Up Log-Down, Extravascular

Parameter	Subject	Value	Reference	Difference	Status
ARsq	1	0.714769	0.714769	2.06501e-14	PASS
ARsq	2	0.990351	0.990351	-2.87548e-14	PASS
ARsq	3	0.776307	0.776307	-2.96763e-13	PASS
ARsq	4	0.837717	0.837717	-1.32561e-13	PASS
ARsq	5	0.82892	0.82892	1.11022e-16	PASS
ARsq	6	0.925179	0.925179	-1.11022e-16	PASS
ARsq	7	0.960416	0.960416	-2.93099e-14	PASS
ARsq	8	0.921954	0.921954	1.11022e-16	PASS
ARsq	9	0.921307	0.921307	0.0	PASS
ARsq	10	0.863912	0.863912	1.19904e-14	PASS
AUCall	1	9566.6	9566.6	0.0	PASS
AUCall	2	10054.3	10054.3	-3.63798e-11	PASS
AUCall	3	5392.46	5392.46	2.72848e-12	PASS
AUCall	4	9297.1	9297.1	-5.45697e-12	PASS
AUCall	5	9519.18	9519.18	-3.63798e-12	PASS
AUCall	6	6948.99	6948.99	-1.81899e-12	PASS
AUCall	7	6988.77	6988.77	9.09495e-13	PASS
AUCall	8	7058.82	7058.82	9.09495e-13	PASS
AUCall	9	8302.37	8302.37	0.0	PASS
AUCall	10	5486.84	5486.84	1.81899e-12	PASS
AUCinf	1	42906.2	42906.2	-3.0559e-10	PASS
AUCinf	2	16097.0	16097.0	-2.91038e-11	PASS
AUCinf	3	26022.1	26022.1	-2.25555e-10	PASS
AUCinf	4	21983.3	21983.3	9.09495e-11	PASS
AUCinf	5	25778.2	25778.2	4.72937e-11	PASS
AUCinf	6	15984.1	15984.1	2.00089e-11	PASS
AUCinf	7	11648.2	11648.2	3.63798e-11	PASS
AUCinf	8	15394.4	15394.4	1.27329e-11	PASS
AUCinf	9	24852.5	24852.5	-3.63798e-12	PASS
AUCinf	10	7941.27	7941.27	1.81899e-12	PASS
AUCinf_pred	1	44223.8	44223.8	-4.43833e-10	PASS
AUCinf_pred	2	15905.3	15905.3	3.09228e-11	PASS
AUCinf_pred	3	25727.4	25727.4	-2.11003e-10	PASS
AUCinf_pred	4	22391.8	22391.8	1.09139e-10	PASS
AUCinf_pred	5	24973.9	24973.9	-2.54659e-11	PASS
AUCinf_pred	6	16301.2	16301.2	-3.63798e-12	PASS
AUCinf_pred	7	11898.7	11898.7	2.72848e-11	PASS
AUCinf_pred	8	15525.1	15525.1	1.81899e-12	PASS
AUCinf_pred	9	24909.5	24909.5	5.82077e-11	PASS
AUCinf_pred	10	7744.63	7744.63	0.0	PASS
AUClast	1	9566.6	9566.6	0.0	PASS
AUClast	2	10054.3	10054.3	-3.63798e-11	PASS
AUClast	3	5392.46	5392.46	2.72848e-12	PASS
AUClast	4	9297.1	9297.1	-5.45697e-12	PASS
AUClast	5	9519.18	9519.18	-3.63798e-12	PASS
AUClast	6	6948.99	6948.99	-1.81899e-12	PASS
AUClast	7	6988.77	6988.77	9.09495e-13	PASS
AUClast	8	7058.82	7058.82	9.09495e-13	PASS
AUClast	9	8302.37	8302.37	0.0	PASS

Parameter	Subject	Value	Reference	Difference	Status
AUClast	10	5486.84	5486.84	1.81899e-12	PASS
AUCpct	1	77.7035	77.7035	-1.84741e-13	PASS
AUCpct	2	37.5395	37.5395	-1.42109e-14	PASS
AUCpct	3	79.2774	79.2774	-1.42109e-13	PASS
AUCpct	4	57.7084	57.7084	8.52651e-14	PASS
AUCpct	5	63.0727	63.0727	-3.55271e-14	PASS
AUCpct	6	56.5258	56.5258	5.68434e-14	PASS
AUCpct	7	40.001	40.001	0.0	PASS
AUCpct	8	54.1467	54.1467	4.26326e-14	PASS
AUCpct	9	66.5935	66.5935	-1.42109e-14	PASS
AUCpct	10	30.9073	30.9073	-3.19744e-14	PASS
AUCtau	1	1268.28	1268.28	-4.3201e-12	PASS
AUCtau	2	1831.82	1831.82	1.36424e-12	PASS
AUCtau	3	754.649	754.649	1.13687e-13	PASS
AUCtau	4	1336.48	1336.48	-3.86535e-12	PASS
AUCtau	5	1310.9	1310.9	1.36424e-12	PASS
AUCtau	6	1114.24	1114.24	-3.41061e-12	PASS
AUCtau	7	1079.37	1079.37	-5.00222e-12	PASS
AUCtau	8	766.62	766.62	-4.54747e-13	PASS
AUCtau	9	1219.63	1219.63	4.54747e-13	PASS
AUCtau	10	970.306	970.306	0.0	PASS
AUMCtau	1	5477.2	5477.2	-2.72848e-12	PASS
AUMCtau	2	8367.57	8367.57	5.45697e-12	PASS
AUMCtau	3	3455.35	3455.35	5.00222e-12	PASS
AUMCtau	4	6014.65	6014.65	3.63798e-12	PASS
AUMCtau	5	6609.79	6609.79	2.72848e-12	PASS
AUMCtau	6	5064.72	5064.72	2.72848e-12	PASS
AUMCtau	7	4976.96	4976.96	9.09495e-13	PASS
AUMCtau	8	2863.01	2863.01	2.27374e-12	PASS
AUMCtau	9	5386.88	5386.88	-2.72848e-12	PASS
AUMCtau	10	4713.48	4713.48	1.81899e-12	PASS
Accind	1	33.3296	33.3296	-4.12115e-13	PASS
Accind	2	8.38727	8.38727	-1.24345e-14	PASS
Accind	3	34.2602	34.2602	-4.05009e-13	PASS
Accind	4	14.9445	14.9445	5.32907e-14	PASS
Accind	5	16.813	16.813	-1.06581e-14	PASS
Accind	6	14.9503	14.9503	-4.26326e-14	PASS
Accind	7	9.42751	9.42751	2.30926e-14	PASS
Accind	8	12.949	12.949	-3.73035e-14	PASS
Accind	9	20.1843	20.1843	-8.52651e-14	PASS
Accind	10	6.97669	6.97669	-3.55271e-15	PASS
Cavg	1	140.92	140.92	1.7053e-13	PASS
Cavg	2	203.536	203.536	1.7053e-13	PASS
Cavg	3	83.8499	83.8499	-4.26326e-14	PASS
Cavg	4	148.498	148.498	-1.13687e-13	PASS
Cavg	5	145.656	145.656	2.55795e-13	PASS
Cavg	6	123.804	123.804	-4.83169e-13	PASS
Cavg	7	119.93	119.93	1.42109e-14	PASS
Cavg	8	85.18	85.18	0.0	PASS
Cavg	9	135.515	135.515	-1.7053e-13	PASS
Cavg	10	107.812	107.812	-1.42109e-14	PASS
Clast	1	112.846	112.846	0.0	PASS
Clast	2	85.241	85.241	0.0	PASS
Clast	3	67.901	67.901	0.0	PASS
Clast	4	97.625	97.625	0.0	PASS
Clast	5	110.778	110.778	0.0	PASS
Clast	6	69.501	69.501	0.0	PASS
Clast	7	58.051	58.051	0.0	PASS

Parameter	Subject	Value	Reference	Difference	Status
Clast	8	74.437	74.437	0.0	PASS
Clast	9	93.44	93.44	0.0	PASS
Clast	10	42.191	42.191	0.0	PASS
Clast_pred	1	117.306	117.306	2.27374e-13	PASS
Clast_pred	2	82.5367	82.5367	-1.13687e-13	PASS
Clast_pred	3	66.9311	66.9311	-1.98952e-13	PASS
Clast_pred	4	100.768	100.768	-7.10543e-14	PASS
Clast_pred	5	105.298	105.298	5.11591e-13	PASS
Clast_pred	6	71.9399	71.9399	2.84217e-14	PASS
Clast_pred	7	61.1727	61.1727	1.27898e-13	PASS
Clast_pred	8	75.6043	75.6043	-2.84217e-14	PASS
Clast_pred	9	93.7618	93.7618	4.26326e-14	PASS
Clast_pred	10	38.8109	38.8109	-3.55271e-14	PASS
Cltau	1	0.0788472	0.0788472	5.55112e-17	PASS
Cltau	2	0.0545905	0.0545905	1.38778e-17	PASS
Cltau	3	0.132512	0.132512	-4.71845e-16	PASS
Cltau	4	0.0748234	0.0748234	-2.77556e-17	PASS
Cltau	5	0.0762832	0.0762832	-2.77556e-17	PASS
Cltau	6	0.0897472	0.0897472	-4.16334e-17	PASS
Cltau	7	0.0926469	0.0926469	-1.38778e-17	PASS
Cltau	8	0.130443	0.130443	-3.05311e-16	PASS
Cltau	9	0.081992	0.081992	0.0	PASS
Cltau	10	0.10306	0.10306	4.02456e-16	PASS
Cmax	1	190.869	190.869	0.0	PASS
Cmax	2	261.177	261.177	0.0	PASS
Cmax	3	105.345	105.345	0.0	PASS
Cmax	4	208.542	208.542	0.0	PASS
Cmax	5	169.334	169.334	0.0	PASS
Cmax	6	154.648	154.648	0.0	PASS
Cmax	7	153.254	153.254	0.0	PASS
Cmax	8	138.327	138.327	0.0	PASS
Cmax	9	167.347	167.347	0.0	PASS
Cmax	10	125.482	125.482	0.0	PASS
Ctau	1	137.722	137.722	0.0	PASS
Ctau	2	204.963	204.963	3.12639e-13	PASS
Ctau	3	84.9156	84.9156	0.0	PASS
Ctau	4	141.997	141.997	-3.97904e-13	PASS
Ctau	5	160.057	160.057	3.97904e-13	PASS
Ctau	6	106.986	106.986	7.10543e-14	PASS
Ctau	7	123.572	123.572	0.0	PASS
Ctau	8	84.9595	84.9595	-1.42109e-14	PASS
Ctau	9	142.566	142.566	0.0	PASS
Ctau	10	122.786	122.786	0.0	PASS
Ctaumin	1	121.239	121.239	0.0	PASS
Ctaumin	2	62.222	62.222	0.0	PASS
Ctaumin	3	49.849	49.849	0.0	PASS
Ctaumin	4	52.421	52.421	0.0	PASS
Ctaumin	5	0.0	0.0	0.0	PASS
Ctaumin	6	57.882	57.882	0.0	PASS
Ctaumin	7	19.95	19.95	0.0	PASS
Ctaumin	8	22.724	22.724	0.0	PASS
Ctaumin	9	105.438	105.438	0.0	PASS
Ctaumin	10	13.634	13.634	0.0	PASS
Dose	1	100.0	100.0	0.0	PASS
Dose	2	100.0	100.0	0.0	PASS
Dose	3	100.0	100.0	0.0	PASS
Dose	4	100.0	100.0	0.0	PASS
Dose	5	100.0	100.0	0.0	PASS

Parameter	Subject	Value	Reference	Difference	Status
Dose	6	100.0	100.0	0.0	PASS
Dose	7	100.0	100.0	0.0	PASS
Dose	8	100.0	100.0	0.0	PASS
Dose	9	100.0	100.0	0.0	PASS
Dose	10	100.0	100.0	0.0	PASS
Fluc	1	49.4112	49.4112	-3.55271e-14	PASS
Fluc	2	97.7495	97.7495	2.84217e-14	PASS
Fluc	3	66.1849	66.1849	-2.84217e-14	PASS
Fluc	4	105.133	105.133	4.54747e-13	PASS
Fluc	5	116.256	116.256	-3.83693e-13	PASS
Fluc	6	78.1603	78.1603	-1.42109e-14	PASS
Fluc	7	111.152	111.152	-2.70006e-13	PASS
Fluc	8	135.716	135.716	-8.52651e-14	PASS
Fluc	9	45.6844	45.6844	1.42109e-14	PASS
Fluc	10	103.744	103.744	1.27898e-13	PASS
Fluctau	1	37.7145	37.7145	-7.10543e-15	PASS
Fluctau	2	27.619	27.619	-7.10543e-15	PASS
Fluctau	3	24.3642	24.3642	1.42109e-14	PASS
Fluctau	4	44.8121	44.8121	3.55271e-14	PASS
Fluctau	5	6.36906	6.36906	0.0	PASS
Fluctau	6	38.4976	38.4976	-7.10543e-15	PASS
Fluctau	7	24.7492	24.7492	-1.06581e-14	PASS
Fluctau	8	62.6526	62.6526	-7.10543e-15	PASS
Fluctau	9	18.2865	18.2865	-1.06581e-14	PASS
Fluctau	10	2.50019	2.50019	4.44089e-15	PASS
HL	1	204.786	204.786	-2.50111e-12	PASS
HL	2	49.1374	49.1374	-4.9738e-14	PASS
HL	3	210.591	210.591	-1.67688e-12	PASS
HL	4	90.0736	90.0736	3.83693e-13	PASS
HL	5	101.734	101.734	-1.42109e-14	PASS
HL	6	90.1095	90.1095	-2.84217e-14	PASS
HL	7	55.6345	55.6345	8.52651e-14	PASS
HL	8	77.6194	77.6194	4.26326e-14	PASS
HL	9	122.771	122.771	-2.41585e-13	PASS
HL	10	40.3233	40.3233	-2.84217e-14	PASS
Kel	1	0.00338474	0.00338474	8.93383e-17	PASS
Kel	2	0.0141063	0.0141063	-1.38778e-17	PASS
Kel	3	0.00329143	0.00329143	5.55112e-17	PASS
Kel	4	0.00769534	0.00769534	5.63785e-17	PASS
Kel	5	0.00681333	0.00681333	8.673620000000001e-18	PASS
Kel	6	0.00769228	0.00769228	6.67869e-17	PASS
Kel	7	0.012459	0.012459	-1.04083e-17	PASS
Kel	8	0.00893008	0.00893008	2.08167e-17	PASS
Kel	9	0.00564586	0.00564586	1.12757e-17	PASS
Kel	10	0.0171897	0.0171897	-6.93889e-18	PASS
LZint	1	5.00849	5.00849	-2.66454e-15	PASS
LZint	2	5.4289	5.4289	-8.88178e-16	PASS
LZint	3	4.44065	4.44065	-4.44089e-15	PASS
LZint	4	5.16688	5.16688	0.0	PASS
LZint	5	5.14736	5.14736	-4.44089e-15	PASS
LZint	6	4.82968	4.82968	-4.44089e-15	PASS
LZint	7	5.01075	5.01075	-3.55271e-15	PASS
LZint	8	4.96848	4.96848	-4.44089e-15	PASS
LZint	9	4.94726	4.94726	8.88178e-16	PASS
LZint	10	4.89636	4.89636	-8.88178e-16	PASS
MRTtauinf	1	299.792	299.792	-2.44427e-12	PASS
MRTtauinf	2	74.655	74.655	-2.84217e-14	PASS
MRTtauinf	3	305.92	305.92	-2.55795e-12	PASS

Parameter	Subject	Value	Reference	Difference	Status
MRTtauinf	4	143.538	143.538	7.10543e-13	PASS
MRTtauinf	5	173.022	173.022	-1.98952e-13	PASS
MRTtauinf	6	124.653	124.653	1.42109e-13	PASS
MRTtauinf	7	92.7359	92.7359	8.52651e-14	PASS
MRTtauinf	8	175.462	175.462	2.84217e-13	PASS
MRTtauinf	9	178.811	178.811	3.41061e-13	PASS
MRTtauinf	10	69.5163	69.5163	-2.84217e-14	PASS
MRTtauinf_pred	1	309.142	309.142	-3.29692e-12	PASS
MRTtauinf_pred	2	73.7131	73.7131	-7.10543e-14	PASS
MRTtauinf_pred	3	302.406	302.406	-3.06954e-12	PASS
MRTtauinf_pred	4	146.289	146.289	2.84217e-14	PASS
MRTtauinf_pred	5	167.5	167.5	-3.97904e-13	PASS
MRTtauinf_pred	6	127.214	127.214	1.56319e-13	PASS
MRTtauinf_pred	7	94.8251	94.8251	1.42109e-13	PASS
MRTtauinf_pred	8	176.996	176.996	3.12639e-13	PASS
MRTtauinf_pred	9	179.231	179.231	2.27374e-13	PASS
MRTtauinf_pred	10	67.6924	67.6924	-2.84217e-14	PASS
NpLZ	1	5.0	5.0	0.0	PASS
NpLZ	2	5.0	5.0	0.0	PASS
NpLZ	3	7.0	7.0	0.0	PASS
NpLZ	4	3.0	3.0	0.0	PASS
NpLZ	5	8.0	8.0	0.0	PASS
NpLZ	6	4.0	4.0	0.0	PASS
NpLZ	7	5.0	5.0	0.0	PASS
NpLZ	8	4.0	4.0	0.0	PASS
NpLZ	9	4.0	4.0	0.0	PASS
NpLZ	10	9.0	9.0	0.0	PASS
Obsnum	1	15.0	15.0	0.0	PASS
Obsnum	2	15.0	15.0	0.0	PASS
Obsnum	3	15.0	15.0	0.0	PASS
Obsnum	4	15.0	15.0	0.0	PASS
Obsnum	5	15.0	15.0	0.0	PASS
Obsnum	6	15.0	15.0	0.0	PASS
Obsnum	7	15.0	15.0	0.0	PASS
Obsnum	8	15.0	15.0	0.0	PASS
Obsnum	9	15.0	15.0	0.0	PASS
Obsnum	10	15.0	15.0	0.0	PASS
Rsq	1	0.786077	0.786077	1.56541e-14	PASS
Rsq	2	0.992764	0.992764	-2.18714e-14	PASS
Rsq	3	0.813589	0.813589	-2.47913e-13	PASS
Rsq	4	0.918859	0.918859	-6.67244e-14	PASS
Rsq	5	0.85336	0.85336	-4.44089e-16	PASS
Rsq	6	0.950119	0.950119	-1.11022e-16	PASS
Rsq	7	0.970312	0.970312	-2.24265e-14	PASS
Rsq	8	0.947969	0.947969	-2.22045e-16	PASS
Rsq	9	0.947538	0.947538	-3.33067e-16	PASS
Rsq	10	0.880923	0.880923	1.0103e-14	PASS
Swing	1	0.57432	0.57432	3.33067e-16	PASS
Swing	2	3.1975	3.1975	5.32907e-15	PASS
Swing	3	1.11328	1.11328	4.21885e-15	PASS
Swing	4	2.97821	2.97821	-1.77636e-15	PASS
Swing	6	1.67178	1.67178	4.44089e-15	PASS
Swing	7	6.6819	6.6819	2.66454e-15	PASS
Swing	8	5.08726	5.08726	-2.66454e-15	PASS
Swing	9	0.58716	0.58716	2.22045e-16	PASS
Swing	10	8.20361	8.20361	-3.55271e-15	PASS
Swingtau	1	0.385902	0.385902	4.44089e-16	PASS
Swingtau	2	0.274267	0.274267	-1.11022e-16	PASS

Parameter	Subject	Value	Reference	Difference	Status
Swingtau	3	0.240584	0.240584	5.55112e-17	PASS
Swingtau	4	0.468637	0.468637	-2.77556e-16	PASS
Swingtau	5	0.0579601	0.0579601	-4.16334e-17	PASS
Swingtau	6	0.445494	0.445494	-1.11022e-16	PASS
Swingtau	7	0.240196	0.240196	3.60822e-16	PASS
Swingtau	8	0.628152	0.628152	0.0	PASS
Swingtau	9	0.17382	0.17382	-3.60822e-16	PASS
Swingtau	10	0.0219527	0.0219527	-3.81639e-17	PASS
Tlag	1	0.0	0.0	0.0	PASS
Tlag	2	0.0	0.0	0.0	PASS
Tlag	3	0.0	0.0	0.0	PASS
Tlag	4	0.0	0.0	0.0	PASS
Tlag	5	0.25	0.25	0.0	PASS
Tlag	6	0.0	0.0	0.0	PASS
Tlag	7	0.0	0.0	0.0	PASS
Tlag	8	0.0	0.0	0.0	PASS
Tlag	9	0.0	0.0	0.0	PASS
Tlag	10	0.0	0.0	0.0	PASS
Tlast	1	72.0	72.0	0.0	PASS
Tlast	2	72.0	72.0	0.0	PASS
Tlast	3	72.0	72.0	0.0	PASS
Tlast	4	72.0	72.0	0.0	PASS
Tlast	5	72.0	72.0	0.0	PASS
Tlast	6	72.0	72.0	0.0	PASS
Tlast	7	72.0	72.0	0.0	PASS
Tlast	8	72.0	72.0	0.0	PASS
Tlast	9	72.0	72.0	0.0	PASS
Tlast	10	72.0	72.0	0.0	PASS
Tmax	1	1.0	1.0	0.0	PASS
Tmax	2	1.0	1.0	0.0	PASS
Tmax	3	1.5	1.5	0.0	PASS
Tmax	4	1.0	1.0	0.0	PASS
Tmax	5	4.0	4.0	0.0	PASS
Tmax	6	2.5	2.5	0.0	PASS
Tmax	7	2.5	2.5	0.0	PASS
Tmax	8	4.0	4.0	0.0	PASS
Tmax	9	3.0	3.0	0.0	PASS
Tmax	10	2.0	2.0	0.0	PASS
Vztau	1	23.2949	23.2949	-1.88294e-13	PASS
Vztau	2	3.86993	3.86993	-7.99361e-15	PASS
Vztau	3	40.2597	40.2597	-4.12115e-13	PASS
Vztau	4	9.7232	9.7232	3.73035e-14	PASS
Vztau	5	11.1962	11.1962	5.32907e-15	PASS
Vztau	6	11.6672	11.6672	-1.42109e-14	PASS
Vztau	7	7.43617	7.43617	9.76996e-15	PASS
Vztau	8	14.6071	14.6071	4.44089e-14	PASS
Vztau	9	14.5225	14.5225	-2.30926e-14	PASS
Vztau	10	5.99545	5.99545	-6.21725e-15	PASS

Comparisons: 329. Outside acceptance criteria: 0. Result: PASS.

4.3.3 Linear-trapezoidal rule; Intravascular; Dosetime 0.0; Tau 12; Dose 120

Code:

```
setdosetime!(ds, DoseTime(dose = 120, time = 0.0, tau = 12))
nca!(ds, adm = :iv, calcm = :lint)
```

Table 4: Plasma data results, Linear-trapezoidal rule, Intravascular

Parameter	Subject	Value	Reference	Difference	Status
ARsq	1	0.714769	0.714769	-3.89242e-9	PASS
ARsq	2	0.990351	0.990351	4.11874e-9	PASS
ARsq	3	0.776307	0.776307	-4.40849e-9	PASS
ARsq	4	0.837717	0.837717	6.63705e-11	PASS
ARsq	5	0.844202	0.844202	2.51363e-9	PASS
ARsq	6	0.925179	0.925179	-3.46941e-10	PASS
ARsq	7	0.960416	0.960416	-4.84619e-10	PASS
ARsq	8	0.921954	0.921954	-2.59589e-10	PASS
ARsq	9	0.921307	0.921307	2.29085e-9	PASS
ARsq	10	0.867669	0.867669	-3.4982e-12	PASS
AUCall	1	9585.42	9585.42	-5.0e-5	PASS
AUCall	2	10112.2	10112.2	-0.0005	PASS
AUCall	3	5396.55	5396.55	-5.0e-5	PASS
AUCall	4	9317.84	9317.84	-5.0e-5	PASS
AUCall	5	9561.26	9561.26	0.0	PASS
AUCall	6	6966.6	6966.6	9.09495e-13	PASS
AUCall	7	7029.57	7029.57	0.0	PASS
AUCall	8	7110.67	7110.67	-9.09495e-13	PASS
AUCall	9	8315.08	8315.08	-5.0e-5	PASS
AUCall	10	5620.89	5620.89	0.0	PASS
AUCinf	1	42925.0	42925.0	7.26088e-5	PASS
AUCinf	2	16154.9	16154.9	0.000134568	PASS
AUCinf	3	26026.2	26026.2	-0.000441279	PASS
AUCinf	4	22004.1	22004.1	-0.000131232	PASS
AUCinf	5	25714.4	25714.4	0.00040248	PASS
AUCinf	6	16001.8	16001.8	-0.000256431	PASS
AUCinf	7	11689.0	11689.0	-0.000326636	PASS
AUCinf	8	15446.2	15446.2	0.000305556	PASS
AUCinf	9	24865.2	24865.2	-3.66208e-5	PASS
AUCinf	10	8171.16	8171.16	6.06491e-6	PASS
AUCinf_pred	1	44242.6	44242.6	0.00026702	PASS
AUCinf_pred	2	15963.2	15963.2	-5.36657e-5	PASS
AUCinf_pred	3	25731.5	25731.5	0.000158813	PASS
AUCinf_pred	4	22412.5	22412.5	4.14188e-7	PASS
AUCinf_pred	5	24900.5	24900.5	8.61188e-5	PASS
AUCinf_pred	6	16318.8	16318.8	0.000318763	PASS
AUCinf_pred	7	11939.5	11939.5	-0.000426398	PASS
AUCinf_pred	8	15576.9	15576.9	0.00020719	PASS
AUCinf_pred	9	24922.2	24922.2	-0.000375061	PASS
AUCinf_pred	10	8002.99	8002.99	1.44416e-5	PASS
AUClast	1	9585.42	9585.42	-5.0e-5	PASS
AUClast	2	10112.2	10112.2	-0.0005	PASS
AUClast	3	5396.55	5396.55	-5.0e-5	PASS
AUClast	4	9317.84	9317.84	-5.0e-5	PASS
AUClast	5	9561.26	9561.26	0.0	PASS
AUClast	6	6966.6	6966.6	9.09495e-13	PASS
AUClast	7	7029.57	7029.57	0.0	PASS
AUClast	8	7110.67	7110.67	-9.09495e-13	PASS
AUClast	9	8315.08	8315.08	-5.0e-5	PASS

Parameter	Subject	Value	Reference	Difference	Status
AUClast	10	5620.89	5620.89	0.0	PASS
AUCpct	1	77.6694	77.6694	-3.34814e-7	PASS
AUCpct	2	37.405	37.405	-3.2005e-7	PASS
AUCpct	3	79.2649	79.2649	9.4124e-8	PASS
AUCpct	4	57.6541	57.654	2.82991e-7	PASS
AUCpct	5	62.8175	62.8175	-6.20524e-8	PASS
AUCpct	6	56.4636	56.4636	-1.88346e-7	PASS
AUCpct	7	39.8614	39.8614	3.80099e-7	PASS
AUCpct	8	53.9649	53.9649	-7.63594e-9	PASS
AUCpct	9	66.5594	66.5594	-7.75843e-8	PASS
AUCpct	10	31.2106	31.2106	-1.10321e-7	PASS
AUCtau	1	1670.1	1670.1	-5.0e-5	PASS
AUCtau	2	2380.27	2380.27	0.0	PASS
AUCtau	3	980.346	980.346	1.13687e-13	PASS
AUCtau	4	1711.04	1711.04	-5.0e-5	PASS
AUCtau	5	1738.46	1738.46	0.0	PASS
AUCtau	6	1410.0	1410.0	0.0	PASS
AUCtau	7	1436.56	1436.56	0.0	PASS
AUCtau	8	1105.07	1105.07	-2.27374e-13	PASS
AUCtau	9	1638.19	1638.19	-5.0e-5	PASS
AUCtau	10	1293.71	1293.71	0.0	PASS
AUMCtau	1	9984.82	9984.82	-5.0e-5	PASS
AUMCtau	2	14630.1	14630.1	-0.0005	PASS
AUMCtau	3	6024.5	6024.5	-5.0e-5	PASS
AUMCtau	4	10299.7	10299.7	-0.00025	PASS
AUMCtau	5	11466.1	11466.1	-0.0005	PASS
AUMCtau	6	8467.36	8467.36	-5.0e-5	PASS
AUMCtau	7	9003.02	9003.02	-5.0e-5	PASS
AUMCtau	8	6457.01	6457.01	-5.0e-5	PASS
AUMCtau	9	10095.8	10095.8	-0.00025	PASS
AUMCtau	10	8367.3	8367.3	0.0	PASS
Accind	1	25.1237	25.1237	-3.62042e-7	PASS
Accind	2	6.42162	6.42162	-2.34987e-8	PASS
Accind	3	25.8216	25.8216	-4.72283e-7	PASS
Accind	4	11.3368	11.3368	-1.75822e-7	PASS
Accind	5	12.6581	12.6581	-1.97611e-7	PASS
Accind	6	11.3411	11.3411	-4.1688e-7	PASS
Accind	7	7.20108	7.20108	4.29193e-8	PASS
Accind	8	9.84069	9.84069	9.75825e-9	PASS
Accind	9	15.2657	15.2657	4.59961e-7	PASS
Accind	10	5.55368	5.55368	3.2638e-8	PASS
Cavg	1	139.175	139.175	-4.16667e-6	PASS
Cavg	2	198.356	198.356	1.66667e-6	PASS
Cavg	3	81.6955	81.6955	1.66667e-7	PASS
Cavg	4	142.586	142.586	2.5e-6	PASS
Cavg	5	144.872	144.872	-3.33333e-6	PASS
Cavg	6	117.5	117.5	3.33333e-6	PASS
Cavg	7	119.713	119.713	1.66667e-6	PASS
Cavg	8	92.0892	92.0892	3.33333e-7	PASS
Cavg	9	136.516	136.516	4.16667e-6	PASS
Cavg	10	107.809	107.809	-5.0e-6	PASS
Cdose	1	0.0	0.0	0.0	PASS
Cdose	2	0.0	0.0	0.0	PASS
Cdose	3	0.0	0.0	0.0	PASS
Cdose	4	0.0	0.0	0.0	PASS
Cdose	5	0.0	0.0	0.0	PASS
Cdose	6	0.0	0.0	0.0	PASS
Cdose	7	0.0	0.0	0.0	PASS

Parameter	Subject	Value	Reference	Difference	Status
Cdose	8	0.0	0.0	0.0	PASS
Cdose	9	0.0	0.0	0.0	PASS
Cdose	10	0.0	0.0	0.0	PASS
Clast	1	112.846	112.846	0.0	PASS
Clast	2	85.241	85.241	0.0	PASS
Clast	3	67.901	67.901	0.0	PASS
Clast	4	97.625	97.625	0.0	PASS
Clast	5	110.778	110.778	0.0	PASS
Clast	6	69.501	69.501	0.0	PASS
Clast	7	58.051	58.051	0.0	PASS
Clast	8	74.437	74.437	0.0	PASS
Clast	9	93.44	93.44	0.0	PASS
Clast	10	42.191	42.191	0.0	PASS
Clast_pred	1	117.306	117.306	-1.09686e-7	PASS
Clast_pred	2	82.5367	82.5367	-1.86741e-7	PASS
Clast_pred	3	66.9311	66.9311	-5.84471e-8	PASS
Clast_pred	4	100.768	100.768	3.50061e-6	PASS
Clast_pred	5	105.196	105.196	3.14119e-6	PASS
Clast_pred	6	71.9399	71.9399	9.56122e-9	PASS
Clast_pred	7	61.1727	61.1727	3.07644e-7	PASS
Clast_pred	8	75.6043	75.6043	-3.56254e-7	PASS
Clast_pred	9	93.7618	93.7618	-4.23203e-7	PASS
Clast_pred	10	39.4088	39.4088	-3.41575e-7	PASS
Cltau	1	0.0718519	0.0718519	-3.78326e-10	PASS
Cltau	2	0.0504145	0.0504145	3.71092e-10	PASS
Cltau	3	0.122406	0.122406	4.07826e-9	PASS
Cltau	4	0.070133	0.070133	-5.97791e-11	PASS
Cltau	5	0.0690266	0.0690266	-2.41938e-10	PASS
Cltau	6	0.0851065	0.0851065	-3.02832e-10	PASS
Cltau	7	0.0835329	0.0835329	1.85984e-10	PASS
Cltau	8	0.10859	0.10859	-3.09517e-9	PASS
Cltau	9	0.0732516	0.0732516	2.56233e-10	PASS
Cltau	10	0.0927567	0.0927567	-3.41832e-11	PASS
Cmax	1	190.869	190.869	0.0	PASS
Cmax	2	261.177	261.177	0.0	PASS
Cmax	3	105.345	105.345	0.0	PASS
Cmax	4	208.542	208.542	0.0	PASS
Cmax	5	169.334	169.334	0.0	PASS
Cmax	6	154.648	154.648	0.0	PASS
Cmax	7	153.254	153.254	0.0	PASS
Cmax	8	138.327	138.327	0.0	PASS
Cmax	9	167.347	167.347	0.0	PASS
Cmax	10	125.482	125.482	0.0	PASS
Ctau	1	144.964	144.964	0.0	PASS
Ctau	2	196.035	196.035	0.0	PASS
Ctau	3	76.027	76.027	0.0	PASS
Ctau	4	132.257	132.257	0.0	PASS
Ctau	5	154.066	154.066	0.0	PASS
Ctau	6	113.751	113.751	0.0	PASS
Ctau	7	123.37	123.37	0.0	PASS
Ctau	8	134.133	134.133	0.0	PASS
Ctau	9	135.58	135.58	0.0	PASS
Ctau	10	106.476	106.476	0.0	PASS
Ctaumin	1	0.0	0.0	0.0	PASS
Ctaumin	2	0.0	0.0	0.0	PASS
Ctaumin	3	0.0	0.0	0.0	PASS
Ctaumin	4	0.0	0.0	0.0	PASS
Ctaumin	5	0.0	0.0	0.0	PASS

Parameter	Subject	Value	Reference	Difference	Status
Ctaumin	6	0.0	0.0	0.0	PASS
Ctaumin	7	0.0	0.0	0.0	PASS
Ctaumin	8	0.0	0.0	0.0	PASS
Ctaumin	9	0.0	0.0	0.0	PASS
Ctaumin	10	0.0	0.0	0.0	PASS
Dose	1	120.0	120.0	0.0	PASS
Dose	2	120.0	120.0	0.0	PASS
Dose	3	120.0	120.0	0.0	PASS
Dose	4	120.0	120.0	0.0	PASS
Dose	5	120.0	120.0	0.0	PASS
Dose	6	120.0	120.0	0.0	PASS
Dose	7	120.0	120.0	0.0	PASS
Dose	8	120.0	120.0	0.0	PASS
Dose	9	120.0	120.0	0.0	PASS
Dose	10	120.0	120.0	0.0	PASS
Fluc	1	137.143	137.143	1.37579e-6	PASS
Fluc	2	131.671	131.671	2.55164e-6	PASS
Fluc	3	128.948	128.948	3.77175e-6	PASS
Fluc	4	146.257	146.257	-4.76688e-6	PASS
Fluc	5	116.886	116.886	-6.32284e-7	PASS
Fluc	6	131.616	131.616	-4.1624e-6	PASS
Fluc	7	128.018	128.018	7.74047e-7	PASS
Fluc	8	150.21	150.21	2.99575e-6	PASS
Fluc	9	122.584	122.584	-3.09065e-6	PASS
Fluc	10	116.393	116.393	4.95355e-6	PASS
Fluctau	1	32.9836	32.9836	1.1183e-7	PASS
Fluctau	2	32.841	32.841	1.23517e-7	PASS
Fluctau	3	35.8869	35.8869	-2.92135e-7	PASS
Fluctau	4	53.5009	53.5009	-2.72453e-7	PASS
Fluctau	5	10.539	10.539	-2.22139e-7	PASS
Fluctau	6	34.806	34.806	-1.82969e-7	PASS
Fluctau	7	24.963	24.963	-2.23501e-7	PASS
Fluctau	8	4.55428	4.55428	-3.14114e-8	PASS
Fluctau	9	23.2698	23.2698	-2.65052e-7	PASS
Fluctau	10	17.6293	17.6293	3.78023e-7	PASS
HL	1	204.786	204.786	-3.0616e-6	PASS
HL	2	49.1374	49.1374	4.37811e-7	PASS
HL	3	210.591	210.591	-3.91935e-6	PASS
HL	4	90.0736	90.0736	1.94606e-8	PASS
HL	5	101.072	101.072	2.23995e-6	PASS
HL	6	90.1095	90.1094	5.7666e-8	PASS
HL	7	55.6345	55.6345	3.82012e-7	PASS
HL	8	77.6194	77.6194	3.08325e-7	PASS
HL	9	122.771	122.771	-5.00549e-7	PASS
HL	10	41.8978	41.8978	1.79993e-8	PASS
Kel	1	0.00338474	0.00338474	4.00078e-11	PASS
Kel	2	0.0141063	0.0141063	-5.67502e-11	PASS
Kel	3	0.00329143	0.00329143	-2.75071e-11	PASS
Kel	4	0.00769534	0.00769534	2.29811e-11	PASS
Kel	5	0.00685799	0.00685799	7.27129e-12	PASS
Kel	6	0.00769228	0.00769228	-3.33622e-11	PASS
Kel	7	0.012459	0.012459	-2.32353e-11	PASS
Kel	8	0.00893008	0.00893008	9.67252e-12	PASS
Kel	9	0.00564586	0.00564586	1.87097e-11	PASS
Kel	10	0.0165438	0.0165438	8.95995e-11	PASS
LZint	1	5.00849	5.00849	-7.44672e-9	PASS
LZint	2	5.4289	5.4289	-4.59704e-9	PASS
LZint	3	4.44065	4.44065	-2.44468e-8	PASS

Parameter	Subject	Value	Reference	Difference	Status
LZint	4	5.16688	5.16688	-3.09526e-8	PASS
LZint	5	5.1496	5.1496	-4.88566e-8	PASS
LZint	6	4.82968	4.82968	4.01706e-8	PASS
LZint	7	5.01075	5.01075	-2.03852e-8	PASS
LZint	8	4.96848	4.96848	-2.75635e-9	PASS
LZint	9	4.94726	4.94726	-1.20523e-8	PASS
LZint	10	4.86514	4.86514	2.37787e-8	PASS
MRTtauinf	1	302.403	302.403	-2.98604e-6	PASS
MRTtauinf	2	75.5906	75.5906	-4.92219e-7	PASS
MRTtauinf	3	312.721	312.721	3.34251e-7	PASS
MRTtauinf	4	148.341	148.341	1.4059e-6	PASS
MRTtauinf	5	172.093	172.093	2.88287e-6	PASS
MRTtauinf	6	130.191	130.191	-1.45103e-6	PASS
MRTtauinf	7	91.9083	91.9083	1.01771e-7	PASS
MRTtauinf	8	161.574	161.574	3.15152e-7	PASS
MRTtauinf	9	176.305	176.305	-4.16405e-6	PASS
MRTtauinf	10	70.2607	70.2607	3.97922e-7	PASS
MRTtauinf_pred	1	311.87	311.87	2.24488e-6	PASS
MRTtauinf_pred	2	74.6241	74.6241	-3.46702e-10	PASS
MRTtauinf_pred	3	309.114	309.114	-4.1848e-6	PASS
MRTtauinf_pred	4	151.205	151.205	2.63821e-6	PASS
MRTtauinf_pred	5	166.475	166.475	-3.3877e-6	PASS
MRTtauinf_pred	6	132.889	132.889	-4.21758e-6	PASS
MRTtauinf_pred	7	94.0013	94.0013	-4.12078e-7	PASS
MRTtauinf_pred	8	162.993	162.993	-3.6208e-6	PASS
MRTtauinf_pred	9	176.722	176.722	1.35238e-6	PASS
MRTtauinf_pred	10	68.7008	68.7008	2.01118e-7	PASS
NpLZ	1	5.0	5.0	0.0	PASS
NpLZ	2	5.0	5.0	0.0	PASS
NpLZ	3	7.0	7.0	0.0	PASS
NpLZ	4	3.0	3.0	0.0	PASS
NpLZ	5	9.0	9.0	0.0	PASS
NpLZ	6	4.0	4.0	0.0	PASS
NpLZ	7	5.0	5.0	0.0	PASS
NpLZ	8	4.0	4.0	0.0	PASS
NpLZ	9	4.0	4.0	0.0	PASS
NpLZ	10	12.0	12.0	0.0	PASS
Obsnum	1	16.0	16.0	0.0	PASS
Obsnum	2	16.0	16.0	0.0	PASS
Obsnum	3	16.0	16.0	0.0	PASS
Obsnum	4	16.0	16.0	0.0	PASS
Obsnum	5	16.0	16.0	0.0	PASS
Obsnum	6	16.0	16.0	0.0	PASS
Obsnum	7	16.0	16.0	0.0	PASS
Obsnum	8	16.0	16.0	0.0	PASS
Obsnum	9	16.0	16.0	0.0	PASS
Obsnum	10	16.0	16.0	0.0	PASS
Rsq	1	0.786077	0.786077	-2.91932e-9	PASS
Rsq	2	0.992764	0.992764	5.89052e-10	PASS
Rsq	3	0.813589	0.813589	-3.40409e-10	PASS
Rsq	4	0.918859	0.918859	-4.96681e-9	PASS
Rsq	5	0.863677	0.863677	-1.55057e-9	PASS
Rsq	6	0.950119	0.950119	-2.31294e-10	PASS
Rsq	7	0.970312	0.970312	4.63654e-9	PASS
Rsq	8	0.947969	0.947969	-1.73059e-10	PASS
Rsq	9	0.947538	0.947538	4.86057e-9	PASS
Rsq	10	0.879699	0.879699	-4.54863e-9	PASS
Swingtau	1	0.316665	0.316665	-2.87051e-9	PASS

Parameter	Subject	Value	Reference	Difference	Status
Swingtau	2	0.332298	0.332298	3.96358e-9	PASS
Swingtau	3	0.385626	0.385626	-8.72322e-10	PASS
Swingtau	4	0.576794	0.576794	-3.12414e-9	PASS
Swingtau	5	0.0991004	0.0991004	-4.5095e-10	PASS
Swingtau	6	0.359531	0.359531	-4.77719e-9	PASS
Swingtau	7	0.242231	0.242231	-1.82621e-9	PASS
Swingtau	8	0.0312675	0.0312675	3.28711e-10	PASS
Swingtau	9	0.234304	0.234304	-3.14206e-10	PASS
Swingtau	10	0.1785	0.1785	-6.79214e-10	PASS
Tlast	1	72.0	72.0	0.0	PASS
Tlast	2	72.0	72.0	0.0	PASS
Tlast	3	72.0	72.0	0.0	PASS
Tlast	4	72.0	72.0	0.0	PASS
Tlast	5	72.0	72.0	0.0	PASS
Tlast	6	72.0	72.0	0.0	PASS
Tlast	7	72.0	72.0	0.0	PASS
Tlast	8	72.0	72.0	0.0	PASS
Tlast	9	72.0	72.0	0.0	PASS
Tlast	10	72.0	72.0	0.0	PASS
Tmax	1	1.0	1.0	0.0	PASS
Tmax	2	1.0	1.0	0.0	PASS
Tmax	3	1.5	1.5	0.0	PASS
Tmax	4	1.0	1.0	0.0	PASS
Tmax	5	4.0	4.0	0.0	PASS
Tmax	6	2.5	2.5	0.0	PASS
Tmax	7	2.5	2.5	0.0	PASS
Tmax	8	4.0	4.0	0.0	PASS
Tmax	9	3.0	3.0	0.0	PASS
Tmax	10	2.0	2.0	0.0	PASS
Vztau	1	21.2282	21.2282	3.23537e-9	PASS
Vztau	2	3.57389	3.57389	3.90072e-8	PASS
Vztau	3	37.1892	37.1892	3.74806e-9	PASS
Vztau	4	9.11369	9.11369	4.34956e-8	PASS
Vztau	5	10.0651	10.0651	-3.89763e-7	PASS
Vztau	6	11.0639	11.0639	3.51787e-7	PASS
Vztau	7	6.70465	6.70465	1.28563e-8	PASS
Vztau	8	12.1601	12.1601	-3.32142e-7	PASS
Vztau	9	12.9744	12.9744	3.98064e-7	PASS
Vztau	10	5.60675	5.60675	2.29662e-8	PASS

Comparisons: 320. Outside acceptance criteria: 0. Result: PASS.

4.3.4 Linear/Log Trapezoidal rule; Extravascular; Dosetime 0.0; Tau 12; Dose 120

Code:

```
setdosetime!(ds, DoseTime(dose = 120, time = 0.0, tau = 12))
nca!(ds, adm = :ev, calcm = :logt)
```

Table 5: Plasma data results, Linear/Log Trapezoidal rule, Extravascular

Parameter	Subject	Value	Reference	Difference	Status
ARsq	1	0.714769	0.714769	-3.89242e-9	PASS
ARsq	2	0.990351	0.990351	4.11874e-9	PASS
ARsq	3	0.776307	0.776307	-4.40849e-9	PASS
ARsq	4	0.837717	0.837717	6.63705e-11	PASS
ARsq	5	0.82892	0.82892	3.67764e-9	PASS
ARsq	6	0.925179	0.925179	-3.46941e-10	PASS
ARsq	7	0.960416	0.960416	-4.84619e-10	PASS
ARsq	8	0.921954	0.921954	-2.59589e-10	PASS
ARsq	9	0.921307	0.921307	2.29085e-9	PASS
ARsq	10	0.863912	0.863912	-4.57725e-9	PASS
AUCall	1	9572.86	9572.86	-4.45335e-6	PASS
AUCall	2	10054.0	10054.0	-0.000233403	PASS
AUCall	3	5391.53	5391.53	4.69342e-5	PASS
AUCall	4	9296.22	9296.22	-2.73472e-5	PASS
AUCall	5	9518.65	9518.65	3.49043e-5	PASS
AUCall	6	6948.58	6948.58	-1.07449e-5	PASS
AUCall	7	6987.06	6987.06	1.49628e-5	PASS
AUCall	8	7064.78	7064.78	-3.50533e-5	PASS
AUCall	9	8298.96	8298.96	-4.81792e-5	PASS
AUCall	10	5485.65	5485.65	-4.49999e-5	PASS
AUCinf	1	42912.5	42912.5	-0.000481845	PASS
AUCinf	2	16096.8	16096.8	0.000401165	PASS
AUCinf	3	26021.2	26021.2	5.56554e-5	PASS
AUCinf	4	21982.5	21982.5	-8.57921e-6	PASS
AUCinf	5	25777.7	25777.7	3.01399e-5	PASS
AUCinf	6	15983.7	15983.7	0.000432824	PASS
AUCinf	7	11646.4	11646.4	-0.000311673	PASS
AUCinf	8	15400.3	15400.3	0.000370503	PASS
AUCinf	9	24849.1	24849.1	6.52e-5	PASS
AUCinf	10	7940.08	7940.08	1.94111e-5	PASS
AUCinf_pred	1	44230.1	44230.1	-0.000287433	PASS
AUCinf_pred	2	15905.1	15905.1	0.000212931	PASS
AUCinf_pred	3	25726.5	25726.5	-0.000344252	PASS
AUCinf_pred	4	22390.9	22390.9	0.000123067	PASS
AUCinf_pred	5	24973.4	24973.4	0.000158343	PASS
AUCinf_pred	6	16300.8	16300.8	8.01813e-6	PASS
AUCinf_pred	7	11897.0	11897.0	-0.000411435	PASS
AUCinf_pred	8	15531.0	15531.0	0.000272137	PASS
AUCinf_pred	9	24906.1	24906.1	-0.00027324	PASS
AUCinf_pred	10	7743.45	7743.45	-3.47449e-5	PASS
AUClast	1	9572.86	9572.86	-4.45335e-6	PASS
AUClast	2	10054.0	10054.0	-0.000233403	PASS
AUClast	3	5391.53	5391.53	4.69342e-5	PASS
AUClast	4	9296.22	9296.22	-2.73472e-5	PASS
AUClast	5	9518.65	9518.65	3.49043e-5	PASS
AUClast	6	6948.58	6948.58	-1.07449e-5	PASS
AUClast	7	6987.06	6987.06	1.49628e-5	PASS

Parameter	Subject	Value	Reference	Difference	Status
AUClast	8	7064.78	7064.78	-3.50533e-5	PASS
AUClast	9	8298.96	8298.96	-4.81792e-5	PASS
AUClast	10	5485.65	5485.65	-4.49999e-5	PASS
AUCpct	1	77.6921	77.6921	6.58264e-8	PASS
AUCpct	2	37.5401	37.5401	-7.83442e-8	PASS
AUCpct	3	79.2802	79.2802	2.66786e-7	PASS
AUCpct	4	57.7107	57.7107	1.31552e-7	PASS
AUCpct	5	63.074	63.074	2.14429e-7	PASS
AUCpct	6	56.5272	56.5272	-2.49981e-7	PASS
AUCpct	7	40.0069	40.0069	-3.705e-7	PASS
AUCpct	8	54.1257	54.1257	4.31309e-7	PASS
AUCpct	9	66.6026	66.6026	-2.93404e-7	PASS
AUCpct	10	30.9119	30.9119	-3.69063e-7	PASS
AUCtau	1	1668.36	1668.36	3.71823e-5	PASS
AUCtau	2	2379.57	2379.57	-6.37262e-7	PASS
AUCtau	3	979.109	979.109	1.92192e-6	PASS
AUCtau	4	1709.79	1709.79	1.65844e-5	PASS
AUCtau	5	1738.24	1738.24	-4.86555e-5	PASS
AUCtau	6	1408.16	1408.16	4.37373e-5	PASS
AUCtau	7	1432.02	1432.02	-4.77703e-5	PASS
AUCtau	8	1080.02	1080.02	-1.60775e-5	PASS
AUCtau	9	1630.98	1630.98	4.46897e-6	PASS
AUCtau	10	1292.83	1292.83	3.3305e-5	PASS
AUMCtau	1	9973.81	9973.81	-4.83851e-5	PASS
AUMCtau	2	14631.1	14631.1	-0.000292668	PASS
AUMCtau	3	6022.93	6022.93	4.13197e-5	PASS
AUMCtau	4	10308.0	10308.0	0.000251193	PASS
AUMCtau	5	11473.1	11473.1	0.000104054	PASS
AUMCtau	6	8471.1	8471.1	3.17542e-5	PASS
AUMCtau	7	8982.04	8982.04	-1.45947e-5	PASS
AUMCtau	8	6271.74	6271.74	1.69278e-5	PASS
AUMCtau	9	10040.8	10040.8	-0.000309414	PASS
AUMCtau	10	8361.79	8361.79	1.16704e-5	PASS
Accind	1	25.1237	25.1237	-3.62042e-7	PASS
Accind	2	6.42162	6.42162	-2.34987e-8	PASS
Accind	3	25.8216	25.8216	-4.72283e-7	PASS
Accind	4	11.3368	11.3368	-1.75822e-7	PASS
Accind	5	12.7377	12.7377	6.70115e-8	PASS
Accind	6	11.3411	11.3411	-4.1688e-7	PASS
Accind	7	7.20108	7.20108	4.29193e-8	PASS
Accind	8	9.84069	9.84069	9.75825e-9	PASS
Accind	9	15.2657	15.2657	4.59961e-7	PASS
Accind	10	5.36503	5.36503	-6.35466e-9	PASS
Cavg	1	139.03	139.03	3.09853e-6	PASS
Cavg	2	198.297	198.297	-3.38644e-6	PASS
Cavg	3	81.5924	81.5924	4.93493e-7	PASS
Cavg	4	142.482	142.482	-1.9513e-6	PASS
Cavg	5	144.853	144.853	9.45378e-7	PASS
Cavg	6	117.346	117.346	-4.68856e-6	PASS
Cavg	7	119.335	119.335	-3.98086e-6	PASS
Cavg	8	90.0019	90.0019	3.26872e-7	PASS
Cavg	9	135.915	135.915	-2.96092e-6	PASS
Cavg	10	107.736	107.736	4.44208e-6	PASS
Clast	1	112.846	112.846	0.0	PASS
Clast	2	85.241	85.241	0.0	PASS
Clast	3	67.901	67.901	0.0	PASS
Clast	4	97.625	97.625	0.0	PASS
Clast	5	110.778	110.778	0.0	PASS

Parameter	Subject	Value	Reference	Difference	Status
Clast	6	69.501	69.501	0.0	PASS
Clast	7	58.051	58.051	0.0	PASS
Clast	8	74.437	74.437	0.0	PASS
Clast	9	93.44	93.44	0.0	PASS
Clast	10	42.191	42.191	0.0	PASS
Clast_pred	1	117.306	117.306	-1.09686e-7	PASS
Clast_pred	2	82.5367	82.5367	-1.86741e-7	PASS
Clast_pred	3	66.9311	66.9311	-5.84471e-8	PASS
Clast_pred	4	100.768	100.768	3.50061e-6	PASS
Clast_pred	5	105.298	105.298	5.80678e-7	PASS
Clast_pred	6	71.9399	71.9399	9.56122e-9	PASS
Clast_pred	7	61.1727	61.1727	3.07644e-7	PASS
Clast_pred	8	75.6043	75.6043	-3.56254e-7	PASS
Clast_pred	9	93.7618	93.7618	-4.23203e-7	PASS
Clast_pred	10	38.8109	38.8109	3.50002e-7	PASS
Cltau	1	0.0719271	0.0719271	-2.83697e-10	PASS
Cltau	2	0.0504294	0.0504294	3.07981e-10	PASS
Cltau	3	0.12256	0.12256	-3.18679e-9	PASS
Cltau	4	0.0701841	0.0701841	3.17018e-10	PASS
Cltau	5	0.0690354	0.0690354	3.1888e-10	PASS
Cltau	6	0.0852177	0.0852178	-3.40351e-10	PASS
Cltau	7	0.0837976	0.0837976	-2.12889e-10	PASS
Cltau	8	0.111109	0.111109	-4.30252e-9	PASS
Cltau	9	0.0735756	0.0735756	-3.69079e-10	PASS
Cltau	10	0.0928198	0.0928198	7.42064e-11	PASS
Cmax	1	190.869	190.869	0.0	PASS
Cmax	2	261.177	261.177	0.0	PASS
Cmax	3	105.345	105.345	0.0	PASS
Cmax	4	208.542	208.542	0.0	PASS
Cmax	5	169.334	169.334	0.0	PASS
Cmax	6	154.648	154.648	0.0	PASS
Cmax	7	153.254	153.254	0.0	PASS
Cmax	8	138.327	138.327	0.0	PASS
Cmax	9	167.347	167.347	0.0	PASS
Cmax	10	125.482	125.482	0.0	PASS
Ctau	1	144.964	144.964	0.0	PASS
Ctau	2	196.035	196.035	0.0	PASS
Ctau	3	76.027	76.027	0.0	PASS
Ctau	4	132.257	132.257	0.0	PASS
Ctau	5	154.066	154.066	0.0	PASS
Ctau	6	113.751	113.751	0.0	PASS
Ctau	7	123.37	123.37	0.0	PASS
Ctau	8	134.133	134.133	0.0	PASS
Ctau	9	135.58	135.58	0.0	PASS
Ctau	10	106.476	106.476	0.0	PASS
Ctaumin	1	0.0	0.0	0.0	PASS
Ctaumin	2	0.0	0.0	0.0	PASS
Ctaumin	3	0.0	0.0	0.0	PASS
Ctaumin	4	0.0	0.0	0.0	PASS
Ctaumin	5	0.0	0.0	0.0	PASS
Ctaumin	6	0.0	0.0	0.0	PASS
Ctaumin	7	0.0	0.0	0.0	PASS
Ctaumin	8	0.0	0.0	0.0	PASS
Ctaumin	9	0.0	0.0	0.0	PASS
Ctaumin	10	0.0	0.0	0.0	PASS
Dose	1	120.0	120.0	0.0	PASS
Dose	2	120.0	120.0	0.0	PASS
Dose	3	120.0	120.0	0.0	PASS

Parameter	Subject	Value	Reference	Difference	Status
Dose	4	120.0	120.0	0.0	PASS
Dose	5	120.0	120.0	0.0	PASS
Dose	6	120.0	120.0	0.0	PASS
Dose	7	120.0	120.0	0.0	PASS
Dose	8	120.0	120.0	0.0	PASS
Dose	9	120.0	120.0	0.0	PASS
Dose	10	120.0	120.0	0.0	PASS
Fluc	1	137.287	137.287	-2.25109e-7	PASS
Fluc	2	131.71	131.71	-3.13436e-6	PASS
Fluc	3	129.111	129.111	2.16088e-6	PASS
Fluc	4	146.363	146.363	4.49786e-6	PASS
Fluc	5	116.9	116.9	4.36295e-6	PASS
Fluc	6	131.788	131.788	-4.50635e-6	PASS
Fluc	7	128.423	128.423	-1.09686e-6	PASS
Fluc	8	153.693	153.693	3.16285e-6	PASS
Fluc	9	123.127	123.127	2.24547e-7	PASS
Fluc	10	116.472	116.472	4.193e-6	PASS
Fluctau	1	33.0181	33.0181	4.28691e-8	PASS
Fluctau	2	32.8507	32.8507	2.90447e-8	PASS
Fluctau	3	35.9323	35.9323	-1.35103e-7	PASS
Fluctau	4	53.54	53.54	-2.19212e-7	PASS
Fluctau	5	10.5403	10.5403	9.66466e-8	PASS
Fluctau	6	34.8515	34.8515	7.83064e-8	PASS
Fluctau	7	25.0421	25.0421	-2.9122e-7	PASS
Fluctau	8	4.6599	4.6599	3.63523e-8	PASS
Fluctau	9	23.3728	23.3728	4.28345e-7	PASS
Fluctau	10	17.6413	17.6413	-3.35856e-7	PASS
HL	1	204.786	204.786	-3.0616e-6	PASS
HL	2	49.1374	49.1374	4.37811e-7	PASS
HL	3	210.591	210.591	-3.91935e-6	PASS
HL	4	90.0736	90.0736	1.94606e-8	PASS
HL	5	101.734	101.734	1.56651e-6	PASS
HL	6	90.1095	90.1094	5.7666e-8	PASS
HL	7	55.6345	55.6345	3.82012e-7	PASS
HL	8	77.6194	77.6194	3.08325e-7	PASS
HL	9	122.771	122.771	-5.00549e-7	PASS
HL	10	40.3233	40.3233	4.40951e-7	PASS
Kel	1	0.00338474	0.00338474	4.00078e-11	PASS
Kel	2	0.0141063	0.0141063	-5.67502e-11	PASS
Kel	3	0.00329143	0.00329143	-2.75071e-11	PASS
Kel	4	0.00769534	0.00769534	2.29811e-11	PASS
Kel	5	0.00681333	0.00681333	1.1549e-11	PASS
Kel	6	0.00769228	0.00769228	-3.33622e-11	PASS
Kel	7	0.012459	0.012459	-2.32353e-11	PASS
Kel	8	0.00893008	0.00893008	9.67252e-12	PASS
Kel	9	0.00564586	0.00564586	1.87097e-11	PASS
Kel	10	0.0171897	0.0171897	-1.6958e-10	PASS
LZint	1	5.00849	5.00849	-7.44672e-9	PASS
LZint	2	5.4289	5.4289	-4.59704e-9	PASS
LZint	3	4.44065	4.44065	-2.44468e-8	PASS
LZint	4	5.16688	5.16688	-3.09526e-8	PASS
LZint	5	5.14736	5.14736	-2.02572e-8	PASS
LZint	6	4.82968	4.82968	4.01706e-8	PASS
LZint	7	5.01075	5.01075	-2.03852e-8	PASS
LZint	8	4.96848	4.96848	-2.75635e-9	PASS
LZint	9	4.94726	4.94726	-1.20523e-8	PASS
LZint	10	4.89636	4.89636	-1.2117e-8	PASS
MRTtauinf	1	302.635	302.635	4.14536e-8	PASS

Parameter	Subject	Value	Reference	Difference	Status
MRTtauinf	2	75.3237	75.3237	-1.85354e-7	PASS
MRTtauinf	3	313.068	313.068	-4.72923e-6	PASS
MRTtauinf	4	148.311	148.311	2.54036e-6	PASS
MRTtauinf	5	172.558	172.558	-4.9523e-6	PASS
MRTtauinf	6	130.226	130.226	4.40834e-6	PASS
MRTtauinf	7	91.8667	91.8667	-1.61179e-7	PASS
MRTtauinf	8	164.918	164.918	3.99825e-6	PASS
MRTtauinf	9	176.985	176.985	1.92764e-6	PASS
MRTtauinf	10	68.1676	68.1676	1.00465e-7	PASS
MRTtauinf_pred	1	312.112	312.112	2.67446e-6	PASS
MRTtauinf_pred	2	74.3569	74.3569	-1.85595e-7	PASS
MRTtauinf_pred	3	309.456	309.456	3.62516e-6	PASS
MRTtauinf_pred	4	151.177	151.177	4.40903e-6	PASS
MRTtauinf_pred	5	167.005	167.005	-2.04977e-6	PASS
MRTtauinf_pred	6	132.927	132.927	-1.15395e-6	PASS
MRTtauinf_pred	7	93.9663	93.9663	-4.66101e-7	PASS
MRTtauinf_pred	8	166.37	166.37	-1.73919e-6	PASS
MRTtauinf_pred	9	177.405	177.405	4.00633e-6	PASS
MRTtauinf_pred	10	66.3424	66.3424	3.70109e-7	PASS
NpLZ	1	5.0	5.0	0.0	PASS
NpLZ	2	5.0	5.0	0.0	PASS
NpLZ	3	7.0	7.0	0.0	PASS
NpLZ	4	3.0	3.0	0.0	PASS
NpLZ	5	8.0	8.0	0.0	PASS
NpLZ	6	4.0	4.0	0.0	PASS
NpLZ	7	5.0	5.0	0.0	PASS
NpLZ	8	4.0	4.0	0.0	PASS
NpLZ	9	4.0	4.0	0.0	PASS
NpLZ	10	9.0	9.0	0.0	PASS
Obsnum	1	16.0	16.0	0.0	PASS
Obsnum	2	16.0	16.0	0.0	PASS
Obsnum	3	16.0	16.0	0.0	PASS
Obsnum	4	16.0	16.0	0.0	PASS
Obsnum	5	16.0	16.0	0.0	PASS
Obsnum	6	16.0	16.0	0.0	PASS
Obsnum	7	16.0	16.0	0.0	PASS
Obsnum	8	16.0	16.0	0.0	PASS
Obsnum	9	16.0	16.0	0.0	PASS
Obsnum	10	16.0	16.0	0.0	PASS
Rsq	1	0.786077	0.786077	-2.91932e-9	PASS
Rsq	2	0.992764	0.992764	5.89052e-10	PASS
Rsq	3	0.813589	0.813589	-3.40409e-10	PASS
Rsq	4	0.918859	0.918859	-4.96681e-9	PASS
Rsq	5	0.85336	0.85336	1.72369e-9	PASS
Rsq	6	0.950119	0.950119	-2.31294e-10	PASS
Rsq	7	0.970312	0.970312	4.63654e-9	PASS
Rsq	8	0.947969	0.947969	-1.73059e-10	PASS
Rsq	9	0.947538	0.947538	4.86057e-9	PASS
Rsq	10	0.880923	0.880923	-2.55092e-10	PASS
Swingtau	1	0.316665	0.316665	-2.87051e-9	PASS
Swingtau	2	0.332298	0.332298	3.96358e-9	PASS
Swingtau	3	0.385626	0.385626	-8.72322e-10	PASS
Swingtau	4	0.576794	0.576794	-3.12414e-9	PASS
Swingtau	5	0.0991004	0.0991004	-4.5095e-10	PASS
Swingtau	6	0.359531	0.359531	-4.77719e-9	PASS
Swingtau	7	0.242231	0.242231	-1.82621e-9	PASS
Swingtau	8	0.0312675	0.0312675	3.28711e-10	PASS
Swingtau	9	0.234304	0.234304	-3.14206e-10	PASS

Parameter	Subject	Value	Reference	Difference	Status
Swingtau	10	0.1785	0.1785	-6.79214e-10	PASS
Tlag	1	0.0	0.0	0.0	PASS
Tlag	2	0.0	0.0	0.0	PASS
Tlag	3	0.0	0.0	0.0	PASS
Tlag	4	0.0	0.0	0.0	PASS
Tlag	5	0.5	0.5	0.0	PASS
Tlag	6	0.0	0.0	0.0	PASS
Tlag	7	0.0	0.0	0.0	PASS
Tlag	8	0.0	0.0	0.0	PASS
Tlag	9	0.0	0.0	0.0	PASS
Tlag	10	0.0	0.0	0.0	PASS
Tlast	1	72.0	72.0	0.0	PASS
Tlast	2	72.0	72.0	0.0	PASS
Tlast	3	72.0	72.0	0.0	PASS
Tlast	4	72.0	72.0	0.0	PASS
Tlast	5	72.0	72.0	0.0	PASS
Tlast	6	72.0	72.0	0.0	PASS
Tlast	7	72.0	72.0	0.0	PASS
Tlast	8	72.0	72.0	0.0	PASS
Tlast	9	72.0	72.0	0.0	PASS
Tlast	10	72.0	72.0	0.0	PASS
Tmax	1	1.0	1.0	0.0	PASS
Tmax	2	1.0	1.0	0.0	PASS
Tmax	3	1.5	1.5	0.0	PASS
Tmax	4	1.0	1.0	0.0	PASS
Tmax	5	4.0	4.0	0.0	PASS
Tmax	6	2.5	2.5	0.0	PASS
Tmax	7	2.5	2.5	0.0	PASS
Tmax	8	4.0	4.0	0.0	PASS
Tmax	9	3.0	3.0	0.0	PASS
Tmax	10	2.0	2.0	0.0	PASS
Vztau	1	21.2504	21.2504	5.59948e-9	PASS
Vztau	2	3.57495	3.57495	3.19325e-8	PASS
Vztau	3	37.2362	37.2362	4.47858e-7	PASS
Vztau	4	9.12034	9.12034	6.663e-9	PASS
Vztau	5	10.1324	10.1324	2.68293e-7	PASS
Vztau	6	11.0783	11.0783	3.7772e-7	PASS
Vztau	7	6.72589	6.72589	9.71119e-10	PASS
Vztau	8	12.4421	12.4421	2.10483e-7	PASS
Vztau	9	13.0318	13.0318	-4.54418e-7	PASS
Vztau	10	5.39972	5.39972	-2.57534e-8	PASS

Comparisons: 320. Outside acceptance criteria: 0. Result: PASS.

4.3.5 Linear-Up Log-Down to Cmax (:luldt); Extravascular; Dosetime 0.0; Tau 12; Dose 120

`:luldt` applies the linear rule up to `Cmax` and the linear-up/log-down rule after it. **Phoenix WinNonlin implements no counterpart**, so no external reference exists for this method and none can be obtained. The values below are an internal baseline: they were produced by this package and are carried forward unchanged, which establishes that the behaviour has not changed and nothing more.

The method itself is qualified in the next chapter, where it is compared with closed-form values and with the two externally qualified methods it is built out of. The baseline here and the analytic qualification there answer different questions, and the report keeps them apart: *has anything moved* against *is it right*.

Code:

```
setdosetime!(ds, DoseTime(dose = 120, time = 0.0, tau = 12))
nca!(ds, adm = :ev, calcm = :luldt)
```

Table 6: Plasma data results, Linear-Up Log-Down to Cmax, Extravascular (internal baseline)

Parameter	Subject	Value	Reference	Difference	Status
AUCinf	1	42913.4	42913.4	0.0	PASS
AUCinf	2	16097.0	16097.0	0.0	PASS
AUCinf	3	26022.1	26022.1	3.63798e-12	PASS
AUCinf	4	21983.3	21983.3	0.0	PASS
AUCinf	5	25778.2	25778.2	0.0	PASS
AUCinf	6	15984.1	15984.1	0.0	PASS
AUCinf	7	11648.3	11648.3	0.0	PASS
AUCinf	8	15408.8	15408.8	0.0	PASS
AUCinf	9	24853.5	24853.5	0.0	PASS
AUCinf	10	7941.27	7941.27	0.0	PASS
AUCinf_pred	1	44231.0	44231.0	0.0	PASS
AUCinf_pred	2	15905.3	15905.3	0.0	PASS
AUCinf_pred	3	25727.4	25727.4	0.0	PASS
AUCinf_pred	4	22391.8	22391.8	0.0	PASS
AUCinf_pred	5	24973.9	24973.9	0.0	PASS
AUCinf_pred	6	16301.2	16301.2	0.0	PASS
AUCinf_pred	7	11898.9	11898.9	-1.81899e-12	PASS
AUCinf_pred	8	15539.6	15539.6	0.0	PASS
AUCinf_pred	9	24910.5	24910.5	0.0	PASS
AUCinf_pred	10	7744.63	7744.63	0.0	PASS
AUClast	1	9573.81	9573.81	0.0	PASS
AUClast	2	10054.3	10054.3	-1.81899e-12	PASS
AUClast	3	5392.46	5392.46	0.0	PASS
AUClast	4	9297.1	9297.1	1.81899e-12	PASS
AUClast	5	9519.18	9519.18	1.81899e-12	PASS
AUClast	6	6948.99	6948.99	0.0	PASS
AUClast	7	6988.96	6988.96	0.0	PASS
AUClast	8	7073.31	7073.31	0.0	PASS
AUClast	9	8303.37	8303.37	0.0	PASS
AUClast	10	5486.84	5486.84	0.0	PASS
Cdose	1	0.0	0.0	0.0	PASS
Cdose	2	0.0	0.0	0.0	PASS
Cdose	3	0.0	0.0	0.0	PASS
Cdose	4	0.0	0.0	0.0	PASS

Parameter	Subject	Value	Reference	Difference	Status
Cdose	5	0.0	0.0	0.0	PASS
Cdose	6	0.0	0.0	0.0	PASS
Cdose	7	0.0	0.0	0.0	PASS
Cdose	8	0.0	0.0	0.0	PASS
Cdose	9	0.0	0.0	0.0	PASS
Cdose	10	0.0	0.0	0.0	PASS
Clast_pred	1	117.306	117.306	-1.09686e-7	PASS
Clast_pred	2	82.5367	82.5367	-1.86741e-7	PASS
Clast_pred	3	66.9311	66.9311	-5.84471e-8	PASS
Clast_pred	4	100.768	100.768	3.50061e-6	PASS
Clast_pred	5	105.298	105.298	5.80678e-7	PASS
Clast_pred	6	71.9399	71.9399	9.56122e-9	PASS
Clast_pred	7	61.1727	61.1727	3.07644e-7	PASS
Clast_pred	8	75.6043	75.6043	-3.56254e-7	PASS
Clast_pred	9	93.7618	93.7618	-4.23203e-7	PASS
Clast_pred	10	38.8109	38.8109	3.50002e-7	PASS
Cmax	1	190.869	190.869	0.0	PASS
Cmax	2	261.177	261.177	0.0	PASS
Cmax	3	105.345	105.345	0.0	PASS
Cmax	4	208.542	208.542	0.0	PASS
Cmax	5	169.334	169.334	0.0	PASS
Cmax	6	154.648	154.648	0.0	PASS
Cmax	7	153.254	153.254	0.0	PASS
Cmax	8	138.327	138.327	0.0	PASS
Cmax	9	167.347	167.347	0.0	PASS
Cmax	10	125.482	125.482	0.0	PASS
Kel	1	0.00338474	0.00338474	4.00078e-11	PASS
Kel	2	0.0141063	0.0141063	-5.67502e-11	PASS
Kel	3	0.00329143	0.00329143	-2.75071e-11	PASS
Kel	4	0.00769534	0.00769534	2.29811e-11	PASS
Kel	5	0.00681333	0.00681333	1.1549e-11	PASS
Kel	6	0.00769228	0.00769228	-3.33622e-11	PASS
Kel	7	0.012459	0.012459	-2.32353e-11	PASS
Kel	8	0.00893008	0.00893008	9.67252e-12	PASS
Kel	9	0.00564586	0.00564586	1.87097e-11	PASS
Kel	10	0.0171897	0.0171897	-1.6958e-10	PASS
Tmax	1	1.0	1.0	0.0	PASS
Tmax	2	1.0	1.0	0.0	PASS
Tmax	3	1.5	1.5	0.0	PASS
Tmax	4	1.0	1.0	0.0	PASS
Tmax	5	4.0	4.0	0.0	PASS
Tmax	6	2.5	2.5	0.0	PASS
Tmax	7	2.5	2.5	0.0	PASS
Tmax	8	4.0	4.0	0.0	PASS
Tmax	9	3.0	3.0	0.0	PASS
Tmax	10	2.0	2.0	0.0	PASS

Comparisons: 80. Outside acceptance criteria: 0. Result: PASS.

4.3.6 Urine data; Linear-trapezoidal rule; Extravascular; Dosetime 0.0; Dose 100

Code:

```
upkds = upkimport(upkdata, :st, :et, :conc, :vol, :subj;
dosetime =  MetidaNCA.DoseTime(dose = 100))
MetidaNCA.nca!(upkds)
```

Table 7: Urine data results

Parameter	Value	Reference	Difference	Status
AR	16.0	16.0	0.0	PASS
ARsq	0.810983	0.810983	1.03367e-9	PASS
AUCall	17.125	17.125	0.0	PASS
AUCinf	19.6042	19.6042	-6.58352e-9	PASS
AUClast	17.125	17.125	0.0	PASS
AUCpct	12.6461	12.6461	1.33044e-7	PASS
HL	5.15526	5.15526	-1.84268e-8	PASS
Kel	0.134454	0.134454	4.59631e-9	PASS
Maxrate	4.0	4.0	0.0	PASS
Prec	16.0	16.0	0.0	PASS
Rlast	0.333333	0.333333	3.33333e-9	PASS
Rsq	0.905492	0.905492	5.16834e-10	PASS
Tmax	1.5	1.5	0.0	PASS
Vol	11.0	11.0	0.0	PASS

Comparisons: 14. Outside acceptance criteria: 0. Result: PASS.

4.3.7 Pharmacodynamics data; Linear-trapezoidal rule

Code:

```
pd =  MetidaNCA.pdimport(pddata, :time, :obs;
bl = 3.0, th = 1.5, id = Dict(:subj => 1))
pdres = MetidaNCA.nca!(pd)
```

Table 8: Pharmacodynamics data results

Parameter	Value	Reference	Difference	Status
AUCABL	7.38571	7.38571	-1.42857e-8	PASS
AUCATH	13.9595	13.9595	-1.90476e-7	PASS
AUCBBL	8.73571	8.73571	-1.42857e-8	PASS
AUCBTH	1.80952	1.80952	9.52381e-9	PASS
AUCNETB	-1.35	-1.35	4.44089e-16	PASS
AUCNETT	12.15	12.15	-1.77636e-15	PASS
BL	3.0	3.0	0.0	PASS
Rmax	8.0	8.0	0.0	PASS
Rmin	0.0	0.0	0.0	PASS
TABL	3.48095	3.48095	-1.90476e-8	PASS
TATH	5.7619	5.7619	-3.80952e-8	PASS
TBBL	5.51905	5.51905	1.90476e-8	PASS
TBTH	3.2381	3.2381	3.80952e-8	PASS
TH	1.5	1.5	0.0	PASS
TIMEBTW	2.28095	2.28095	-1.90476e-8	PASS
Tmax	5.0	5.0	0.0	PASS

Comparisons: 16. Outside acceptance criteria: 0. Result: PASS.

5 Analytic qualification of the integration rules

The four integration rules are composed of two formulas, the linear and the logarithmic trapezoid, together with a rule for choosing between them on each interval. Both components are qualified in this chapter without a second implementation. For `:luldt` no second implementation is available: Phoenix WinNonlin offers no counterpart, so the preceding chapter establishes a baseline only.

The qualification rests on an exact property. The logarithmic trapezoid is exact on a mono-exponential profile, and the linear trapezoid is exact on piecewise-linear data. Where the property holds, a closed form is a reference independent of any implementation. Four blocks follow.

- On a mono-exponential profile the three rules that use the logarithmic formula reproduce the closed-form integral, and the `nca!` chain reproduces the closed forms for `AUClast`, `AUMClast`, `AUCinf`, `AUMCinf`, `Kel`, `HL` and `MRTinf`.
- On a strictly ascending profile the logarithmic formula is applicable to none of the rules, so all four return the area of the linear trapezoids.
- The switching rule is established by exact identities with `:logt` and `:luld`, both qualified externally in the preceding chapter. `:luldt` differs from `:logt` on an ascending interval after `Cmax` and from `:luld` on a descending interval before `Cmax`. Four profiles are chosen so that each identity and each difference is exercised.
- The rules are rewritten directly from the formulas and compared with the package over every subject of the validation data set.

The mono-exponential intravenous-bolus profile of section 5.2 is the profile with analytically known values that a qualification data set is required to contain (chapter *Validation coverage*).

5.1 The two formulas and the direct reimplementation

```
# Written from the formulas, with no reference to the internals of the package.
lintrap(t₁, t₂, c₁, c₂) = (t₂ - t₁) * (c₁ + c₂) / 2
logtrap(t₁, t₂, c₁, c₂) = (t₂ - t₁) * (c₁ - c₂) / log(c₁ / c₂)

function trapauc(time, obs, rule)
    tmaxn = argmax(obs)                          # first maximum, as in Cmax
    s = zero(eltype(obs)) * zero(eltype(time))
    for i in 1:length(time) - 1
        t₁, t₂, c₁, c₂ = time[i], time[i+1], obs[i], obs[i+1]
        after = i >= tmaxn
        uselog =
            rule === :lint  ? false :
            rule === :logt  ? (after && c₁ > 0 && c₂ > 0) :
            rule === :luld  ? (c₁ > c₂ > 0) :
            rule === :luldt ? (after && c₁ > c₂ > 0) :
            error("Unknown rule: $rule")
        s += uselog ? logtrap(t₁, t₂, c₁, c₂) : lintrap(t₁, t₂, c₁, c₂)
    end
    return s
```

```
end
```

```
trapauc (generic function with 1 method)
```

5.2 Mono-exponential profile against closed forms

A profile generated by $C(t) = C_0 e^{-kt}$ with $C_0 = 100$ and $k = 0.35$ h [1], sampled at 0, 0.5, 1, 2, 3.5, 5, 8 and 12 h. The references are

$$AUC_{0\to t} = \frac{C_0 - C_{last}}{k}, \qquad AUMC_{0\to t} = -\frac{t_{last} C_{last}}{k} + \frac{C_0 - C_{last}}{k^2},$$

$$AUC_{0\to\infty} = \frac{C_0}{k}, \qquad AUMC_{0\to\infty} = \frac{C_0}{k^2}$$

and $MRT_{0\to\infty} = 1/k$, $HL = \ln 2/k$, $\lambda_z = k$, $R^2 = 1$.

Table 9: Mono-exponential profile, computed values against closed forms

Parameter	Rule	Value	Closed form	Difference	Status
AUClast (auc)	logt	281.42984	281.42984	0.0	PASS
Rsq	logt	1.0	1.0	2.66e-15	PASS
Kel	logt	0.35	0.35	3.33e-16	PASS
HL	logt	1.9804205	1.9804205	-2.0e-15	PASS
AUClast	logt	281.42984	281.42984	0.0	PASS
AUMClast	logt	752.67184	752.67184	0.0	PASS
AUCinf	logt	285.71429	285.71429	0.0	PASS
AUMCinf	logt	816.32653	816.32653	0.0	PASS
MRTinf	logt	2.8571429	2.8571429	4.44e-16	PASS
AUClast (auc)	luld	281.42984	281.42984	0.0	PASS
Rsq	luld	1.0	1.0	2.66e-15	PASS
Kel	luld	0.35	0.35	3.33e-16	PASS
HL	luld	1.9804205	1.9804205	-2.0e-15	PASS
AUClast	luld	281.42984	281.42984	0.0	PASS
AUMClast	luld	752.67184	752.67184	0.0	PASS
AUCinf	luld	285.71429	285.71429	0.0	PASS
AUMCinf	luld	816.32653	816.32653	0.0	PASS
MRTinf	luld	2.8571429	2.8571429	4.44e-16	PASS
AUClast (auc)	luldt	281.42984	281.42984	0.0	PASS
Rsq	luldt	1.0	1.0	2.66e-15	PASS
Kel	luldt	0.35	0.35	3.33e-16	PASS
HL	luldt	1.9804205	1.9804205	-2.0e-15	PASS
AUClast	luldt	281.42984	281.42984	0.0	PASS
AUMClast	luldt	752.67184	752.67184	0.0	PASS
AUCinf	luldt	285.71429	285.71429	0.0	PASS
AUMCinf	luldt	816.32653	816.32653	0.0	PASS
MRTinf	luldt	2.8571429	2.8571429	4.44e-16	PASS

Comparisons: 27. Outside acceptance criteria: 0. Result: PASS.

5.3 Strictly ascending profile

With concentrations rising at every interval the logarithmic formula is applicable to none of the rules, so all four must return the same number: the area of the linear trapezoids.

Table 10: Strictly ascending profile, all four rules against the linear trapezoids

Rule	Value	Linear trapezoids	Difference	Status
lint	497.5	497.5	0.0	PASS
logt	497.5	497.5	0.0	PASS
luld	497.5	497.5	0.0	PASS
luldt	497.5	497.5	0.0	PASS

Comparisons: 4. Outside acceptance criteria: 0. Result: PASS.

5.4 Switching-rule identities

`:logt` and `:luld` are qualified externally. `:luldt` is pinned relative to them: it must agree with `:logt` except on an ascending interval after `Cmax`, and with `:luld` except on a descending interval before `Cmax`. Profile B contains the first case, profile D the second, and for those two profiles the agreement is required to break. A `:luldt` that coincided with `:logt` and with `:luld` on every profile would not implement the switching rule.

Table 11: Switching-rule identities of `:luldt` against the externally qualified rules

Profile	Against	Required	:luldt	Other rule	Status
A single peak	:logt	equal	404.20209	404.20209	PASS
A single peak	:luld	equal	404.20209	404.20209	PASS
B rise after Cmax	:logt	differ	448.02956	447.35563	PASS
B rise after Cmax	:luld	equal	448.02956	448.02956	PASS
C monotone decline	:logt	equal	280.2367	280.2367	PASS
C monotone decline	:luld	equal	280.2367	280.2367	PASS
D dip before Cmax	:logt	equal	522.24993	522.24993	PASS
D dip before Cmax	:luld	differ	522.24993	520.58097	PASS

Comparisons: 6. Outside acceptance criteria: 0. Result: PASS.

5.5 Direct reimplementation on the validation data set

The four rules as written above, against the package, over every subject of the plasma data set. This is a second implementation of the same formulas, so it catches an error in the package's own loop structure that a closed form on a smooth profile would not.

Table 12: Direct reimplementation of the four rules against the package, by rule

Rule	Subjects	Max. difference	Status
lint	10	0.0	PASS
logt	10	0.0	PASS
luld	10	0.0	PASS
luldt	10	0.0	PASS

Comparisons: 40. Outside acceptance criteria: 0. Result: PASS.

6 Derived parameters

Six parameters have no external reference. `Cllast` and `Vzlast` relate the dose to the observed area, not to the extrapolated area. `Vsstau` is the steady-state volume over the dosing interval. `LZ` is the regression slope with its sign. `Rsqn` is the index of the candidate interval selected by the terminal-phase search. `AUCBTW` is the area between the baseline and the threshold. Phoenix WinNonlin publishes none of the six.

For each of the six the report establishes that the computed value equals the value given by the identity that defines the parameter, where the parameters on the right-hand side are themselves qualified. The evidence is of the kind described as derived in section 1.2.1. A derived comparison inherits whatever its inputs inherit and establishes nothing about the inputs themselves. The reference is the right-hand side computed from the same result, so the comparison requires exact agreement.

`Minrate` is documented as the lowest excretion rate of a urine profile and is not qualified in this chapter. The computed value does not meet the documented definition: `nca!` prepends a zero to the excretion-rate vector to give the curve an origin and computes `Minrate` after the insertion, so the value is identically zero. Two data sets were examined, with observed excretion rates of 0.333 to 4.0 and of 75 to 450. `Minrate` equals 0.0 in both. The coverage table reports `Minrate` as not covered and states the reason.

Table 13: Derived parameters against the identities that define them

Parameter	Identity	Subject	Value	Identity gives	Status
LZ	-Kel	1	-0.0033847439	-0.0033847439	PASS
Cllast	Dose / AUClast	1	0.010432509	0.010432509	PASS
Vzlast	Dose / (AUClast * Kel)	1	3.0822152	3.0822152	PASS
Vssinf	Clinf * MRTinf	1	0.68296357	0.68296357	PASS
Rsqn	ARsq = keldata[Rsqn]	1	0.71476928	0.71476928	PASS
LZ	-Kel	2	-0.014106315	-0.014106315	PASS
Cllast	Dose / AUClast	2	0.0098890689	0.0098890689	PASS
Vzlast	Dose / (AUClast * Kel)	2	0.70103843	0.70103843	PASS
Vssinf	Clinf * MRTinf	2	0.44530008	0.44530008	PASS
Rsqn	ARsq = keldata[Rsqn]	2	0.99035145	0.99035145	PASS
LZ	-Kel	3	-0.0032914304	-0.0032914304	PASS
Cllast	Dose / AUClast	3	0.018530358	0.018530358	PASS
Vzlast	Dose / (AUClast * Kel)	3	5.6298801	5.6298801	PASS
Vssinf	Clinf * MRTinf	3	1.1720533	1.1720533	PASS
Rsqn	ARsq = keldata[Rsqn]	3	0.77630678	0.77630678	PASS
LZ	-Kel	4	-0.0076953442	-0.0076953442	PASS
Cllast	Dose / AUClast	4	0.010732106	0.010732106	PASS
Vzlast	Dose / (AUClast * Kel)	4	1.3946232	1.3946232	PASS
Vssinf	Clinf * MRTinf	4	0.59397938	0.59397938	PASS
Rsqn	ARsq = keldata[Rsqn]	4	0.83771737	0.83771737	PASS
LZ	-Kel	5	-0.0068133279	-0.0068133279	PASS
Cllast	Dose / AUClast	5	0.010458873	0.010458873	PASS

Parameter	Identity	Subject	Value	Identity gives	Status
Vzlast	Dose / (AUClast * Kel)	5	1.5350608	1.5350608	PASS
Vssinf	Clinf * MRTinf	5	0.58081039	0.58081039	PASS
Rsqn	ARsq = keldata[Rsqn]	5	0.82891994	0.82891994	PASS
LZ	-Kel	6	-0.0076922807	-0.0076922807	PASS
Cllast	Dose / AUClast	6	0.014354208	0.014354208	PASS
Vzlast	Dose / (AUClast * Kel)	6	1.8660537	1.8660537	PASS
Vssinf	Clinf * MRTinf	6	0.80141896	0.80141896	PASS
Rsqn	ARsq = keldata[Rsqn]	6	0.92517856	0.92517856	PASS
LZ	-Kel	7	-0.012458956	-0.012458956	PASS
Cllast	Dose / AUClast	7	0.014225614	0.014225614	PASS
Vzlast	Dose / (AUClast * Kel)	7	1.1417982	1.1417982	PASS
Vssinf	Clinf * MRTinf	7	0.68011442	0.68011442	PASS
Rsqn	ARsq = keldata[Rsqn]	7	0.96041642	0.96041642	PASS
LZ	-Kel	8	-0.0089300798	-0.0089300798	PASS
Cllast	Dose / AUClast	8	0.014063363	0.014063363	PASS
Vzlast	Dose / (AUClast * Kel)	8	1.5748307	1.5748307	PASS
Vssinf	Clinf * MRTinf	8	0.74359406	0.74359406	PASS
Rsqn	ARsq = keldata[Rsqn]	8	0.92195356	0.92195356	PASS
LZ	-Kel	9	-0.0056458649	-0.0056458649	PASS
Cllast	Dose / AUClast	9	0.012026342	0.012026342	PASS
Vzlast	Dose / (AUClast * Kel)	9	2.1301151	2.1301151	PASS
Vssinf	Clinf * MRTinf	9	0.71174887	0.71174887	PASS
Rsqn	ARsq = keldata[Rsqn]	9	0.92130684	0.92130684	PASS
LZ	-Kel	10	-0.017189737	-0.017189737	PASS
Cllast	Dose / AUClast	10	0.017790763	0.017790763	PASS
Vzlast	Dose / (AUClast * Kel)	10	1.0349642	1.0349642	PASS
Vssinf	Clinf * MRTinf	10	0.72748096	0.72748096	PASS
Rsqn	ARsq = keldata[Rsqn]	10	0.86391165	0.86391165	PASS
Vsstau	Cltau * MRTtauinf	1	21.728235	21.728235	PASS
MRTinf_pred	AUMCinf_pred / AUCinf_pred	1	295.37445	295.37445	PASS
Vsstau	Cltau * MRTtauinf	2	3.8108592	3.8108592	PASS
MRTinf_pred	AUMCinf_pred / AUCinf_pred	2	71.08582	71.08582	PASS
Vsstau	Cltau * MRTtauinf	3	38.278842	38.278842	PASS
MRTinf_pred	AUMCinf_pred / AUCinf_pred	3	304.23015	304.23015	PASS
Vsstau	Cltau * MRTtauinf	4	10.403572	10.403572	PASS
MRTinf_pred	AUMCinf_pred / AUCinf_pred	4	131.99805	131.99805	PASS
Vsstau	Cltau * MRTtauinf	5	11.929466	11.929466	PASS

Parameter	Identity	Subject	Value	Identity gives	Status
MRTinf_pred	AUMCinf_pred / AUCinf_pred	5	147.7548	147.7548	PASS
Vsstau	Cltau * MRTtauinf	6	11.080068	11.080068	PASS
MRTinf_pred	AUMCinf_pred / AUCinf_pred	6	129.67423	129.67423	PASS
Vsstau	Cltau * MRTtauinf	7	7.6773678	7.6773678	PASS
MRTinf_pred	AUMCinf_pred / AUCinf_pred	7	81.025282	81.025282	PASS
Vsstau	Cltau * MRTtauinf	8	17.545381	17.545381	PASS
MRTinf_pred	AUMCinf_pred / AUCinf_pred	8	115.43715	115.43715	PASS
Vsstau	Cltau * MRTtauinf	9	12.914588	12.914588	PASS
MRTinf_pred	AUMCinf_pred / AUCinf_pred	9	177.14308	177.14308	PASS
Vsstau	Cltau * MRTtauinf	10	6.4346996	6.4346996	PASS
MRTinf_pred	AUMCinf_pred / AUCinf_pred	10	56.963742	56.963742	PASS
AUCBTW	AUCBBL - AUCBTH	-	6.9261905	6.9261905	PASS

Comparisons: 71. Outside acceptance criteria: 0. Result: PASS.

7 Graphics

Plotting belongs to the validated scope. A figure is a result that reaches the report, and a figure built from the wrong data reports the wrong result. The failure mode is in the data handed to the drawing code: the wrong subject, the wrong column, or a profile reordered on its way to the axes.

Each figure of this chapter is reproduced for visual verification. The data the figure carries is then read back out of the plot object and compared with the data the package computed. A figure carries the computed numbers without rounding them, so the comparison requires exact agreement.

7.1 Figures

Seven figures, one per plotting entry point of the package. Each group of figures is followed by `\clearpage`. Captioned figures are floats, and a sequence of floats with no intervening text exhausts the float queue of the typesetter and stops the build, so the queue is flushed after each group.

7.1.1 Individual profile

The profile of one subject on a linear and on a semi-logarithmic axis. The semi-logarithmic axis is the axis on which the terminal phase is assessed visually. Non-positive observations cannot be placed on a logarithmic axis and are dropped. The number of dropped observations is compared with the number of non-positive observations in the table below.

```
dsp   = MetidaNCA.pkimport(pkdata2, :Time, :Concentration, [:Subject, :Formulation];
                           dosetime = MetidaNCA.DoseTime(dose = 100, time = 0.0))
dspr  = MetidaNCA.nca!(dsp)
pkplot(dsp[1])
```

```
pkplot(dsp[1]; yscale = :log10)
```

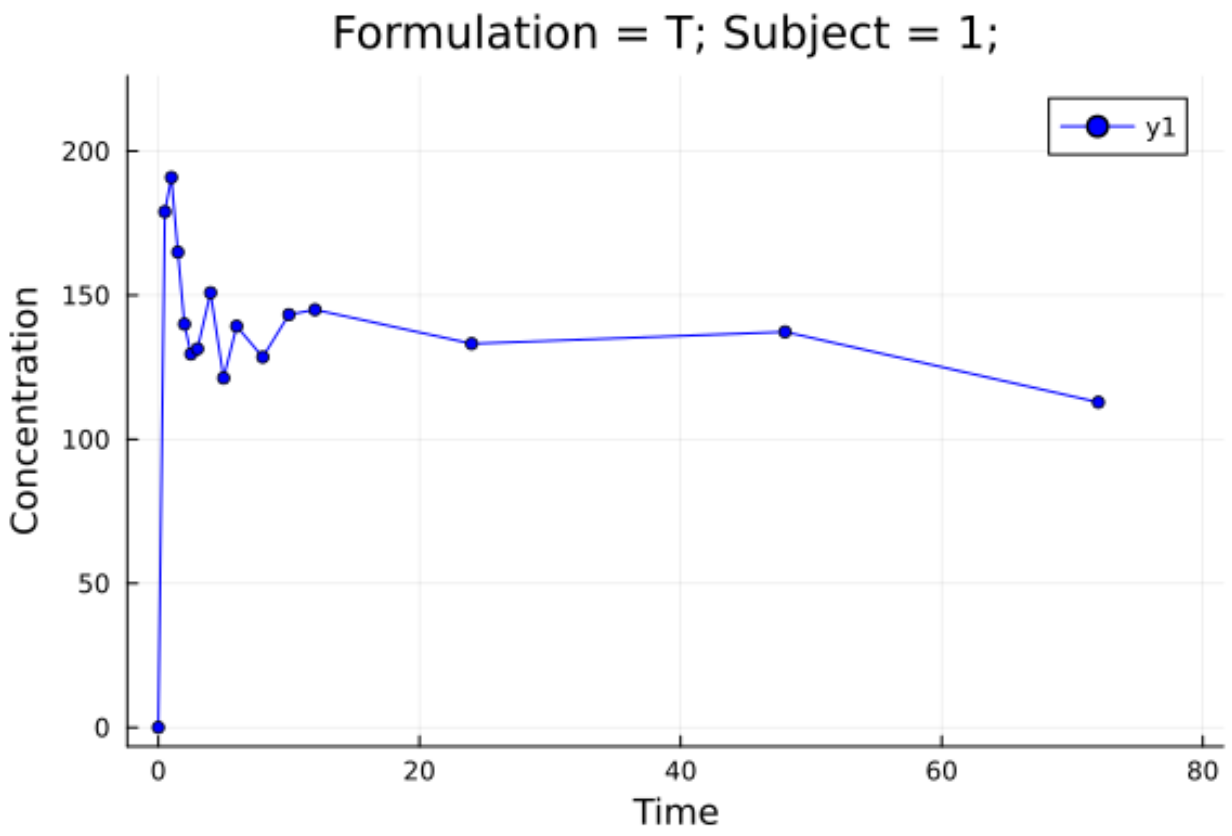


Figure 1: Individual profile, linear scale

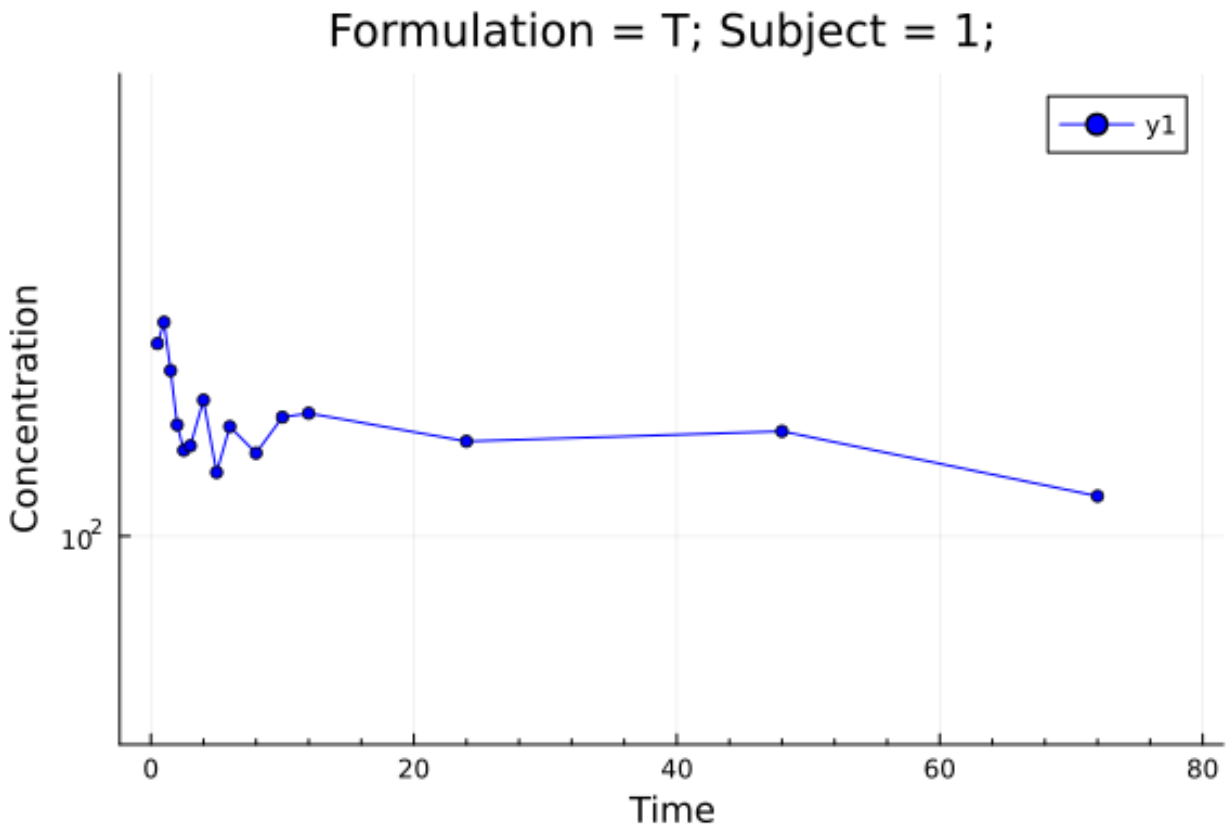


Figure 2: Individual profile, semi-logarithmic scale

7.1.2 Data set and mean profile

Every subject of the data set on one pair of axes, followed by the mean profile with its interval. The overlay is the figure on which a misordered data set is visible. The check below compares all ten series with all ten subjects.

```
pkplot(dsp; typesort = :Subject, pagesort = MetidaNCA.NoPageSort())
```

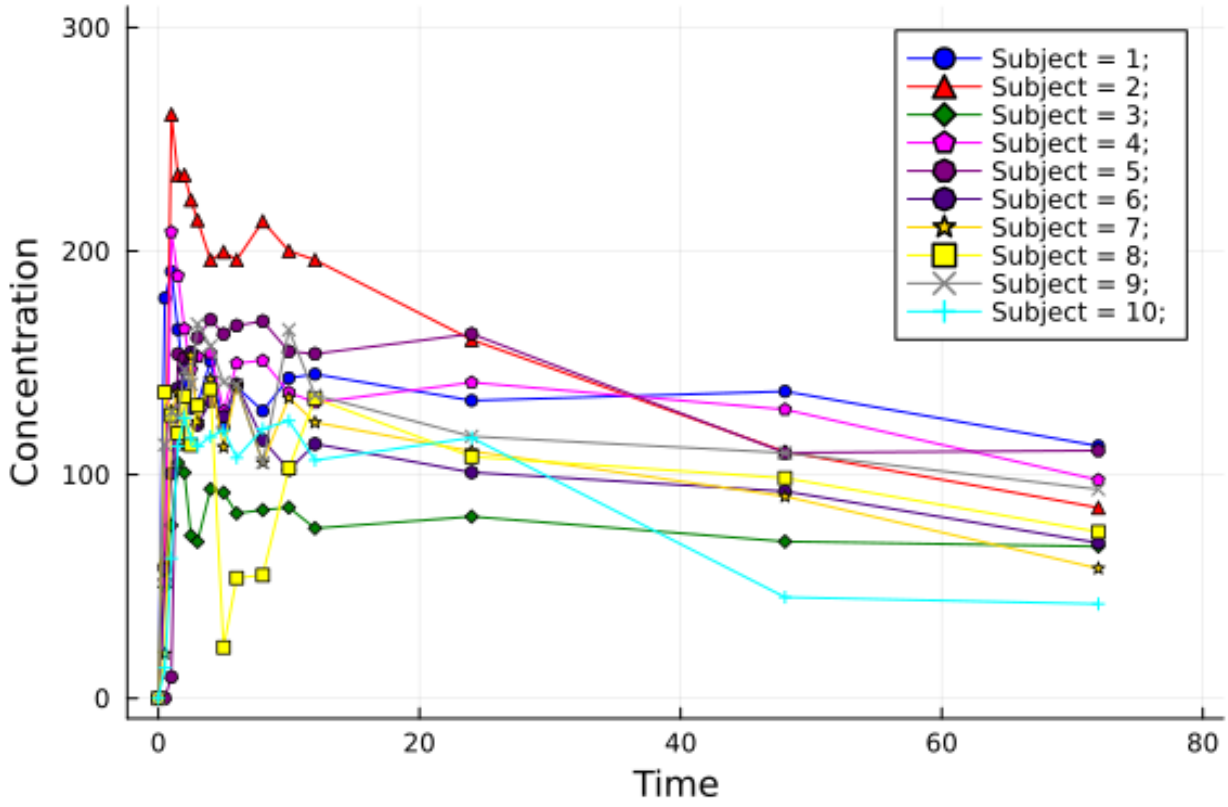


Figure 3: Data set, one series per subject

```
vpcplot(dsp)
```

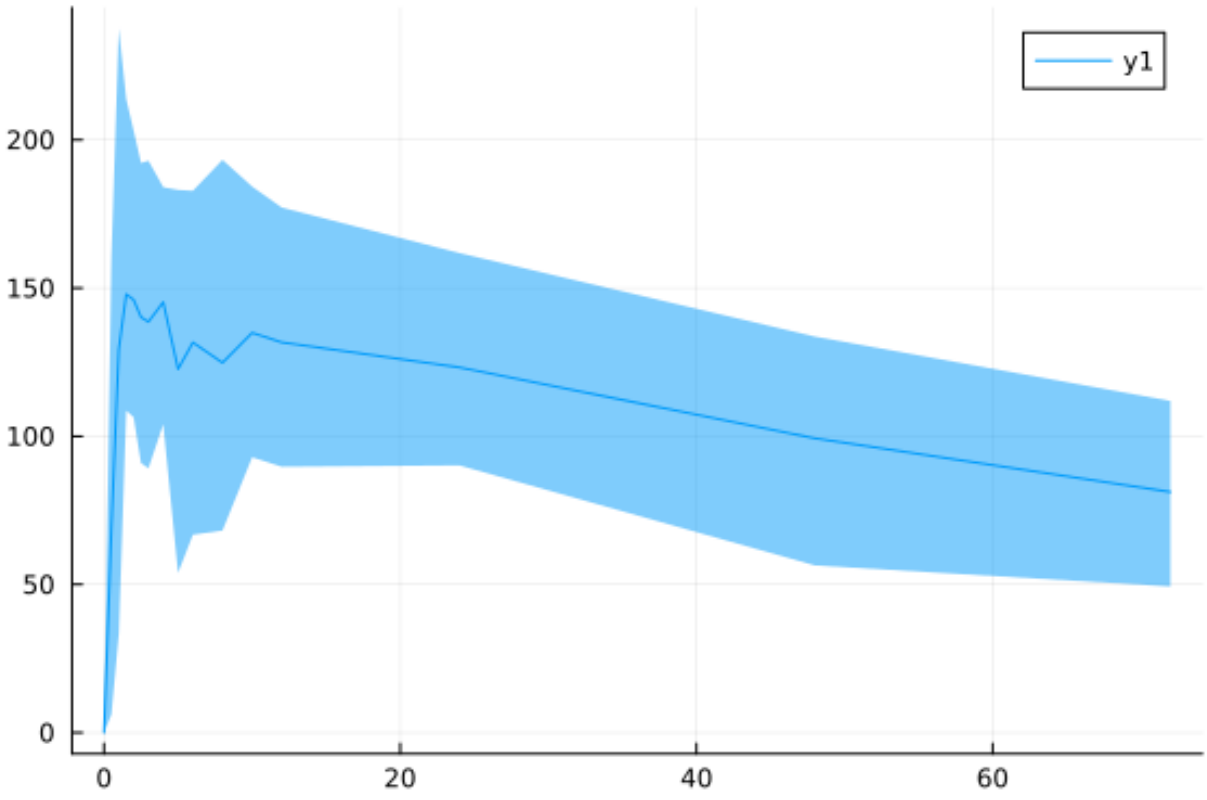


Figure 4: Mean profile with interval

7.1.3 Terminal phase and pharmacodynamics

The terminal-phase figure draws the fitted line over the profile. The number of points of the fitted line is compared with `NpLZ`, the number of points the computation used. The comparison establishes that the line of the figure is fitted to the interval from which the reported parameters were obtained.

```
pkplot(dspr[1]; elim = true)
```

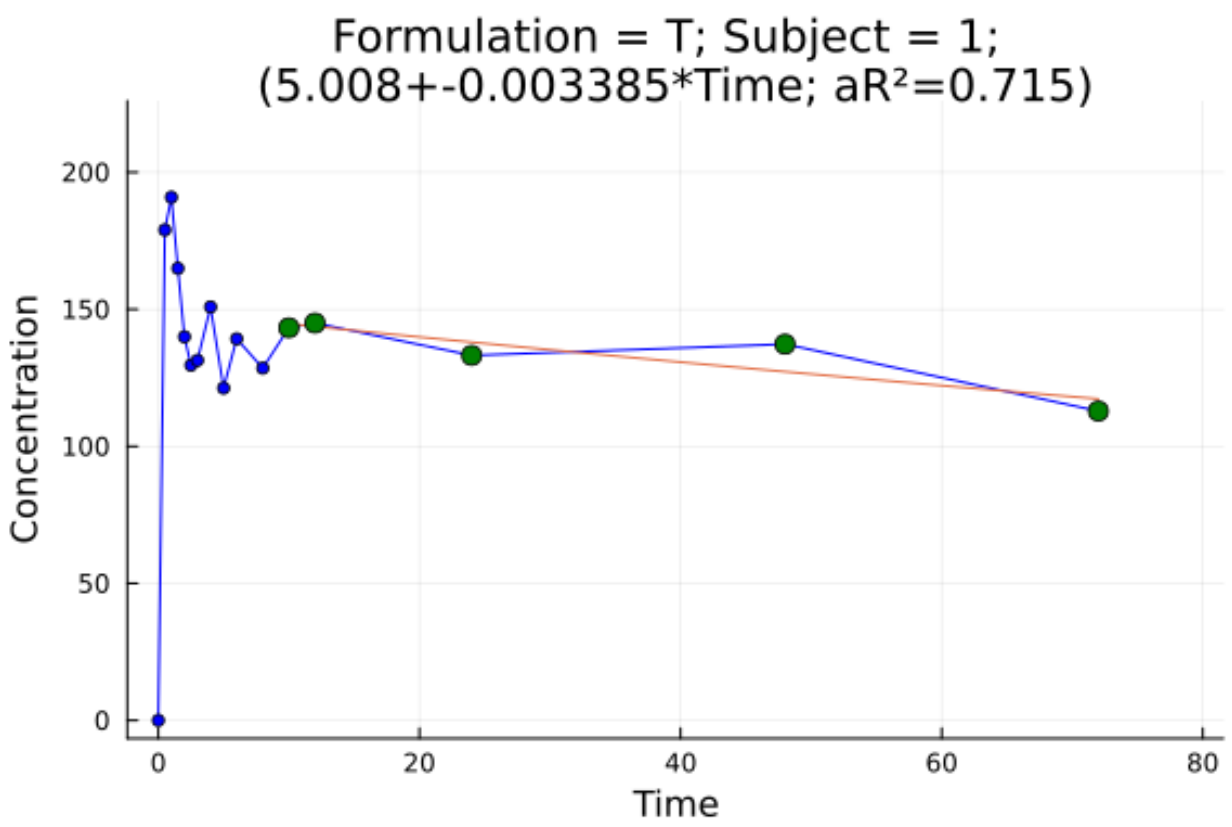


Figure 5: Terminal-phase interval and the points used for it

```
pkplot(pdp; drawbl = true, drawth = true)
```

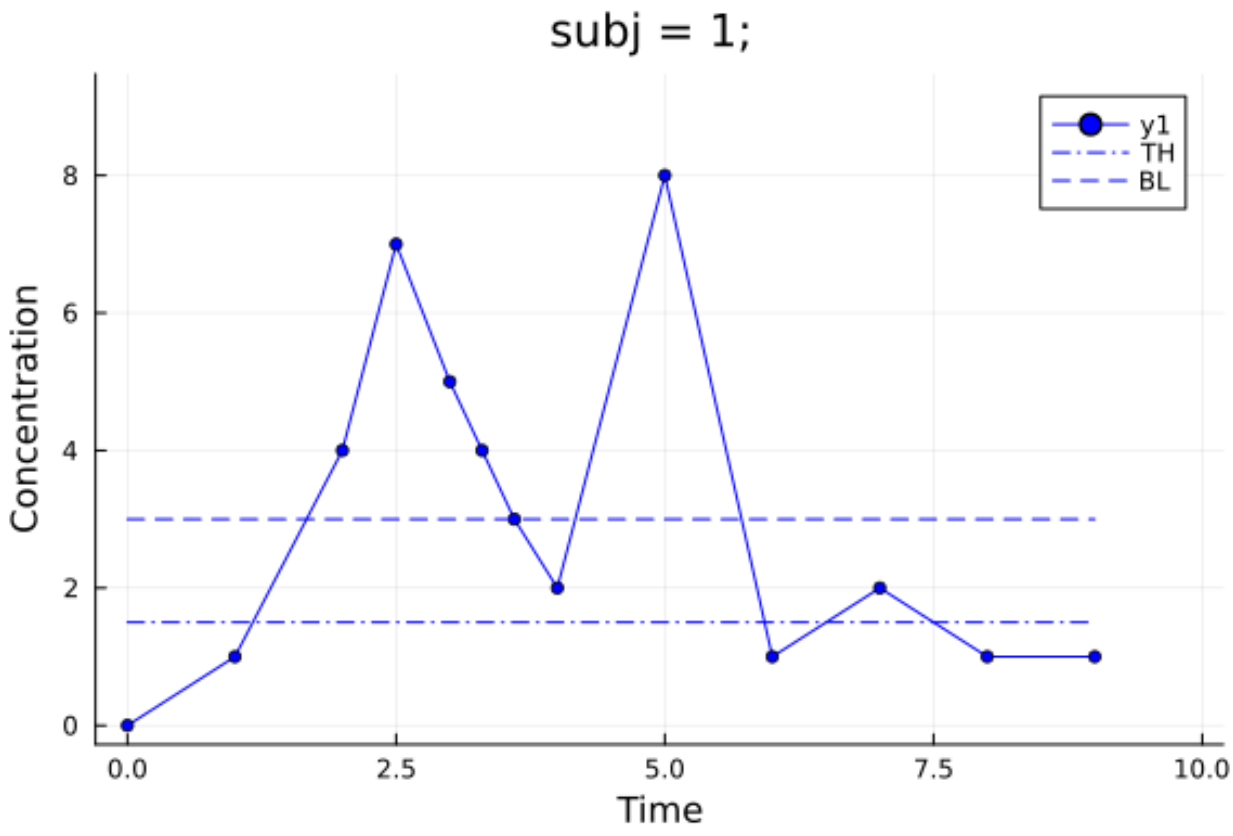


Figure 6: Pharmacodynamics profile with baseline and threshold

7.2 The figures carry the computed data

Table 14: Data held by each figure against the data computed by the package

Figure	Checked	Value	Reference	Status
Individual, linear	series count	1	1	PASS
Individual, linear	time, max. difference	0.0	0	PASS
Individual, linear	observations, max. difference	0.0	0	PASS
Individual, semi-log	series count	1	1	PASS
Individual, semi-log	time, max. difference	0.0	0	PASS
Individual, semi-log	observations, max. difference	0.0	0	PASS
Individual, semi-log	points dropped	1	1	PASS
Data set	series count	10	10	PASS
Data set	all subjects, max. difference	0.0	0	PASS
Mean profile	time, max. difference	0.0	0	PASS
Mean profile	mean, max. difference	0.0	0	PASS
Terminal phase	series count, minimum	2	2	PASS
Terminal phase	profile obs., max. difference	0.0	0	PASS
Terminal phase	points in the fit	5	5.0	PASS
Pharmacodynamics	observations, max. difference	0.0	0	PASS
Pharmacodynamics	series count, minimum	3	3	PASS

7.3 Urine excretion rate

A urine subject is not drawn the way a plasma subject is drawn. Its observations are concentrations in a collected volume, and its time field holds collection intervals. Neither quantity is placed on an axis in that form.

The quantity a urine profile has as a curve is the excretion rate, $C \cdot Vol/(t_{end} - t_{start})$. The time coordinate of the excretion rate is the midpoint of the interval over which the sample was collected. The midpoint is the coordinate the analysis uses for the excretion rate and for the regression of $\ln Rate$ that gives λ_z (chapter *Performance qualification*, urine scenario). Use of the start or the end of the interval shifts the curve, and with it `TmaxRate`, by up to half a collection interval.

`MetidaNCA.midtime` returns the midpoints and `MetidaNCA.excretionrate` returns the excretion rates. The figure is drawn from both.

```
pkplot(upkp[1])
```

Table 15: The urine figure against the computed excretion rate

Figure	Checked	Value	Reference	Status
Urine	series count	1	1	PASS
Urine	midpoints, max. difference	0.0	0	PASS
Urine	rate, max. difference	0.0	0	PASS
Urine	time of the maximum	1.5	1.5	PASS
Urine	height of the maximum	4.0	4.0	PASS

Comparisons: 21. Outside acceptance criteria: 0. Result: PASS.

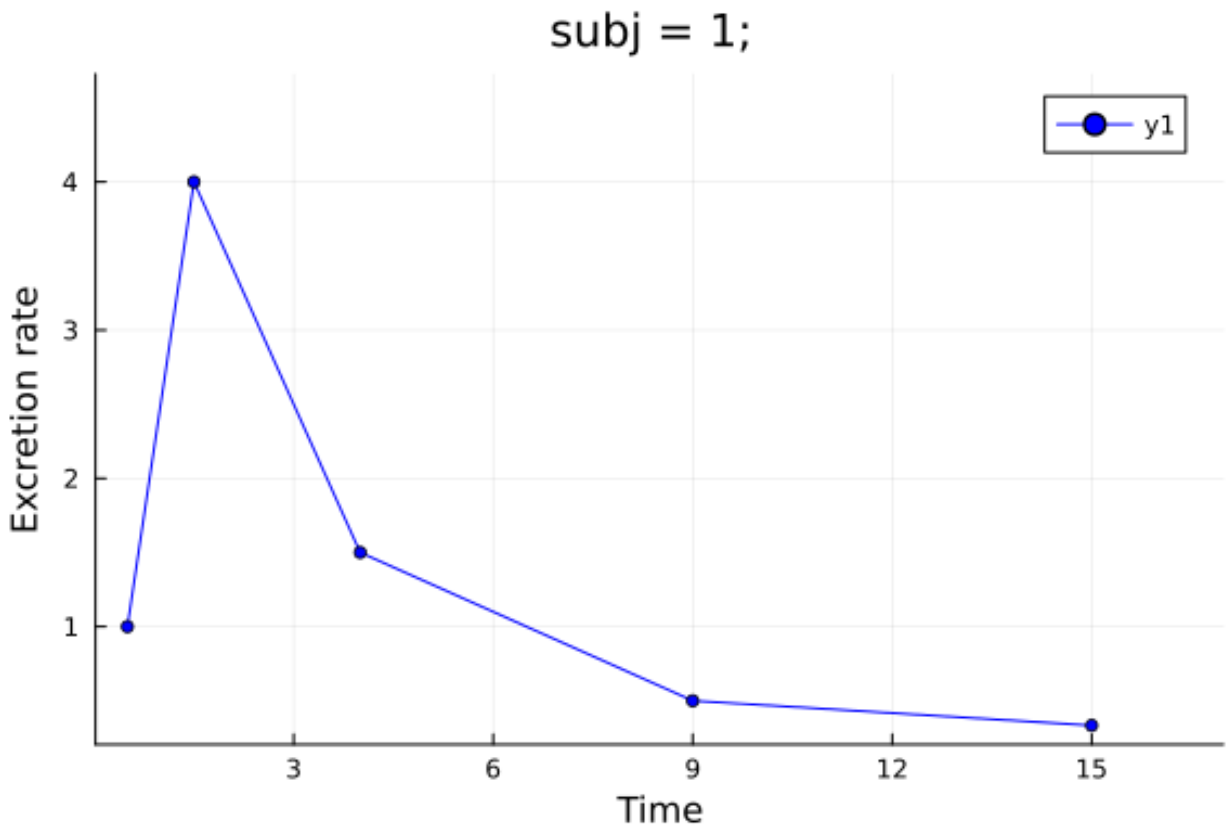


Figure 7: Urine excretion rate against the midpoint of each collection interval

8 Validation coverage

The chapter states what the report establishes and what it does not establish. Both tables are built from the acceptance log that produced the verdict. A parameter the package begins to produce is reported as NOT COVERED until a scenario compares it, and a capability whose scenario did not report is shown with that status.

The kinds of evidence are defined in section 1.2.1 and are not interchangeable.

Kind	What it establishes	Source of the reference
external	agreement with an independent implementation	Phoenix WinNonlin 8.0, Appendix 2
analytic	agreement with a closed form	chapter *Analytic qualification of the integration rules*
derived	agreement with the identity that defines the parameter	chapter *Derived parameters*
internal	that the behaviour has not changed	a value fixed by this package

An internal baseline is not evidence of correctness. It is used only where no external reference can be obtained, and the report states the fact at the point of use.

8.1 Parameters

Table 17: Coverage of every parameter produced during this build

Parameter	External	Analytic	Derived	Internal	Checks	Status
AR	yes	-	-	-	1	qualified
ARsq	yes	-	-	-	41	qualified
AUCABL	yes	-	-	-	1	qualified
AUCATH	yes	-	-	-	1	qualified
AUCBBL	yes	-	-	-	1	qualified
AUCBTH	yes	-	-	-	1	qualified
AUCBTW	-	-	yes	-	1	derived
AUCNETB	yes	-	-	-	1	qualified
AUCNETT	yes	-	-	-	1	qualified
AUCall	yes	-	-	-	41	qualified
AUCinf	yes	yes	-	yes	54	qualified
AUCinf_pred	yes	-	-	yes	50	qualified
AUClast	yes	yes	-	yes	107	qualified
AUCpct	yes	-	-	-	41	qualified
AUCtau	yes	-	-	-	30	qualified
AUMCinf	yes	yes	-	-	13	qualified
AUMCinf_pred	yes	-	-	-	10	qualified
AUMClast	yes	yes	-	-	13	qualified
AUMCtau	yes	-	-	-	30	qualified
Accind	yes	-	-	-	30	qualified
BL	yes	-	-	-	1	qualified
Cavg	yes	-	-	-	30	qualified
Cdose	yes	-	-	yes	20	qualified
Clast	yes	-	-	-	40	qualified
Clast_pred	yes	-	-	yes	50	qualified
Clinf	yes	-	-	-	10	qualified
Cllast	-	-	yes	-	10	derived
Cltau	yes	-	-	-	30	qualified
Cmax	yes	-	-	yes	50	qualified
Ctau	yes	-	-	-	30	qualified
Ctaumin	yes	-	-	-	30	qualified
Dose	yes	-	-	-	40	qualified
Fluc	yes	-	-	-	30	qualified
Fluctau	yes	-	-	-	30	qualified
HL	yes	yes	-	-	44	qualified
Kel	yes	yes	-	yes	54	qualified
LZ	-	-	yes	-	10	derived
LZint	yes	-	-	-	40	qualified
MRTinf	yes	yes	-	-	13	qualified
MRTinf_pred	yes	-	yes	-	20	qualified
MRTlast	yes	-	-	-	10	qualified
MRTtauinf	yes	-	-	-	30	qualified
MRTtauinf_pred	yes	-	-	-	30	qualified
Maxrate	yes	-	-	-	1	qualified
Minrate	-	-	-	-	0	NOT COVERED
NpLZ	yes	-	-	-	40	qualified
Obsnum	yes	-	-	-	40	qualified
Prec	yes	-	-	-	1	qualified
Rlast	yes	-	-	-	1	qualified
Rmax	yes	-	-	-	1	qualified
Rmin	yes	-	-	-	1	qualified
Rsq	yes	yes	-	-	44	qualified
Rsqn	-	-	yes	-	10	derived
Swing	yes	-	-	-	9	qualified
Swingtau	yes	-	-	-	30	qualified

Parameter	External	Analytic	Derived	Internal	Checks	Status
TABL	yes	-	-	-	1	qualified
TATH	yes	-	-	-	1	qualified
TBBL	yes	-	-	-	1	qualified
TBTH	yes	-	-	-	1	qualified
TH	yes	-	-	-	1	qualified
TIMEBTW	yes	-	-	-	1	qualified
Tlag	yes	-	-	-	30	qualified
Tlast	yes	-	-	-	40	qualified
Tmax	yes	-	-	yes	52	qualified
Vol	yes	-	-	-	1	qualified
Vssinf	-	-	yes	-	10	derived
Vsstau	-	-	yes	-	10	derived
Vzinf	yes	-	-	-	10	qualified
Vzlast	-	-	yes	-	10	derived
Vztau	yes	-	-	-	30	qualified

Parameters produced or checked during this build: 70. Qualified against an external or analytic reference: 62. Established through a defining identity only: 7. Resting on an internal regression baseline only: 0. Not covered by any comparison in this report: 1.

Identity only: AUCBTW, Cllast, LZ, Rsqn, Vssinf, Vsstau, Vzlast. No other program reports these parameters. For each of them the report establishes that the computed value equals the value given by its defining identity, computed from parameters that are themselves qualified (chapter *Derived parameters*).

Not covered: Minrate. The package computes these parameters and reports them to the user. No comparison in this report establishes their values. Where the package test suite exercises them, the coverage is operational and is not a numeric qualification.

8.2 Functionality

The table states the declared scope of the validation. The *Scenarios* column names the blocks of this report that exercise each capability and is filled from the acceptance log. The *Evidence* column states the kind of evidence those blocks provide. Where this report does not cover a capability, the *Evidence* column states where the capability is covered instead.

Table 18: Coverage of the package functionality by this report

Functionality	Scenarios	Evidence	Note
Plasma data import (pkimport)	PK-LINT-EV-DT0, PK-LULD-EV-DT025-TAU9	external	Every plasma scenario imports through it
Urine data import (upkimport)	URINE-LINT	external	
Pharmacodynamics import (pdimport)	PD-LINT	external	
calcm = :lint	PK-LINT-EV-DT0, PK-LINT-IV-TAU12, TRAP-ASC, TRAP-REIMP	analytic, external	
calcm = :logt	PK-LOGT-EV-TAU12, TRAP-EXP, TRAP-REIMP	analytic, external	
calcm = :luld	PK-LULD-EV-DT025-TAU9, TRAP-EXP, TRAP-REIMP	analytic, external	
calcm = :luldt	PK-LULDT-EV-TAU12, TRAP-EXP, TRAP-ID, TRAP-REIMP	analytic, internal	No external counterpart exists. Qualified analytically, see the integration-rules chapter
adm = :ev	PK-LINT-EV-DT0, PK-LOGT-EV-TAU12	external	
adm = :iv	PK-LINT-IV-TAU12, TRAP-EXP	analytic, external	
Infusion (DoseTime rate, IT)	-	not implemented	DoseTime.rate is stored and printed but enters no computation. MRT carries no IT/2 correction
Steady state (tau): AUCtau, Ctau, Cavg, Fluc, Swing, Accind	PK-LULD-EV-DT025-TAU9, PK-LINT-IV-TAU12, PK-LOGT-EV-TAU12	external	
Dose time other than zero	PK-LULD-EV-DT025-TAU9	external	
Terminal phase: automatic interval search	PK-LINT-EV-DT0, TRAP-EXP	analytic, external	Kel, HL, Rsq, ARsq, NpLZ compared in every plasma scenario
Terminal phase: user-set interval (ElimRange, setkelrange!, kelexcl)	-	package test suite	Exercised by the test suite. No reference comparison in this report
Partial AUC (partials, prtext)	-	package test suite	Exercised by the test suite; no reference comparison in this report
BLOQ and missing values (LimitRule, applylimitrule!)	-	package test suite	Exercised by the test suite; no reference comparison in this report
Sparse data (auc_sparse)	-	package test suite	Formula checked by the test suite. Standard errors are not implemented

Functionality	Scenarios	Evidence	Note
Multiple dose times (usedose)	-	package test suite	
Multiple observation columns	-	package test suite	
Units (Unitful quantities)	-	package test suite	
Export: DataFrame, metida_table	-	package test suite	Row and subject order checked by the test suite
Graphics: individual profile, linear and semi-logarithmic	GRAPH-DATA	operational	Figures reproduced below. The data they carry is compared with the computed data
Graphics: data set overlay, mean profile	GRAPH-DATA	operational	
Graphics: terminal-phase interval and excluded points	GRAPH-DATA	operational	
Graphics: pharmacodynamics with baseline and threshold	GRAPH-DATA	operational	
Graphics: urine excretion rate against interval midpoints	GRAPH-DATA	operational	The figure is checked against midtime and excretionrate, and its maximum against TmaxRate and MaxRate
Minrate (urine)	-	cannot be qualified	Identically zero: nca! prepends a zero to the rate vector for the curve origin and takes the minimum after that (nca.jl, urine branch)
Derived parameters (Cllast, Vzlast, Vsstau, LZ, Rsqn, AUCBTW)	DERIVED-ID	derived	No external counterpart exists. Each is checked against the identity that defines it
Descriptive statistics (n, mean, SD, CV, geometric mean and CV, median, range)	-	not implemented	The package computes individual parameters only
Subsetting: timefilter, subset	-	package test suite	

Capabilities in the declared scope: 30. Qualified in this report: 18. Exercised by the package test suite only: 9. Not implemented, not supported, or not validated anywhere: 3.

A capability exercised by the test suite is covered operationally. The code runs and its result is self-consistent. Operational coverage establishes less than a comparison with an external or an analytic reference, and the table reports the two separately.

9 Conclusion

9.1 Verdict

Result: PASS. 1518 comparisons over 14 scenarios, all within the acceptance criteria (rtol = 0.0001, atol = 0.0). The package test suite passed. Every required scenario reported. The divergence journal below holds no rows.

9.2 Summary by scenario

The table lists every scenario of this report with the number of comparisons it made against reference values (*Checks*) and the number of recorded outcomes that failed (*Failures*).

- For the plasma, urine and pharmacodynamics scenarios a failure is a comparison outside the acceptance criteria.
- `PKG-TEST` makes no numeric comparison. Its single recorded outcome is the result of the package test suite, so its *Checks* value is zero, and a failure of `PKG-TEST` means the test suite failed.
- A required scenario whose code chunk errored out produces no rows and is listed as NOT REPORTED. The results table of such a scenario is also absent from the Results chapter.

The integration method is named in the table by its `calcm` value (`:lint`, `:luld`, `:logt`, `:luldt`). The abbreviation `dt` denotes the dose time. The Results chapter states each scenario in full.

Table 19: Comparisons and verdict by scenario

Scenario	Description	Checks	Failures	Status
PKG-TEST	Package test suite	0	0	PASS
PK-LINT-EV-DT0	Plasma :lint, EV, dt 0, no tau, dose 100	270	0	PASS
PK-LULD-EV-DT025-TAU9	Plasma :luld, EV, dt 0.25, tau 9, dose 100	329	0	PASS
PK-LINT-IV-TAU12	Plasma :lint, IV, dt 0, tau 12, dose 120	320	0	PASS
PK-LOGT-EV-TAU12	Plasma :logt, EV, dt 0, tau 12, dose 120	320	0	PASS
PK-LULDT-EV-TAU12	Plasma :luldt, EV, dt 0, tau 12, dose 120	80	0	PASS
URINE-LINT	Urine :lint, dose 100	14	0	PASS
PD-LINT	PD :lint, BL 3.0, TH 1.5	16	0	PASS
TRAP-EXP	Integration rules on a mono-exponential profile	27	0	PASS
TRAP-ASC	Integration rules on a strictly ascending profile	4	0	PASS
TRAP-ID	Switching-rule identities of :luldt	6	0	PASS
TRAP-REIMP	Integration rules against a direct reimplementation	40	0	PASS
DERIVED-ID	Derived parameters against their defining identities	71	0	PASS
GRAPH-DATA	Figures carry the computed data	21	0	PASS

9.3 Divergence journal

The journal lists every recorded outcome that failed acceptance, one row per outcome. Three kinds of outcome appear: a comparison outside the criteria stated in section *Acceptance criteria* ($r_{tol} = 0.0001$, $a_{tol} = 0.0$), a failed test suite run, and a required scenario that did not report. For a comparison the row carries the computed value, the reference value, their difference and the relative error $|x - x_{ref}|/|x_{ref}|$.

No divergences. Every recorded comparison is within the acceptance criteria. The package test suite passed. Every required scenario reported. The journal file `docs/src/validation_divergences.tsv` is written with its header and no rows.

Full comparison log: docs/src/validation_acceptance.tsv

Divergence journal: docs/src/validation_divergences.tsv

10 Glossary

- Installation qualification (IQ) - Establishing confidence that process equipment and ancillary systems are compliant with appropriate codes and approved design intentions, and that manufacturer's recommendations are suitably considered.
- Operational qualification (OQ) Establishing confidence that process equipment and sub-systems are capable of consistently operating within established limits and tolerances.
- Product performance qualification (PQ) - Establishing confidence through appropriate testing that the finished product produced by a specified process meets all release requirements for functionality and safety.
- Repository - GitHub repository: https://github.com/PharmCat/MetidaNCA.jl
- Master branch - main branch on GitHub (link).
- Current machine - pc that used for validation report generating.

11 Reference

- General Principles of Software Validation; Final Guidance for Industry and FDA Staff
- Guidance for Industry Process Validation: General Principles and Practices
- Glossary of Computer System Software Development Terminology

12 Appendix 1

12.0.1 Testing PK dataset.

Subject	Formulation	Time	Concentration
1	T	0.0	0.0
1	T	0.5	178.949
1	T	1.0	190.869
1	T	1.5	164.927
1	T	2.0	139.962
1	T	2.5	129.59
1	T	3.0	131.369
1	T	4.0	150.854
1	T	5.0	121.239
1	T	6.0	139.229
1	T	8.0	128.52
1	T	10.0	143.243
1	T	12.0	144.964
1	T	24.0	133.16
1	T	48.0	137.271
1	T	72.0	112.846
2	R	0.0	0.0
2	R	0.5	62.222
2	R	1.0	261.177
2	R	1.5	234.063
2	R	2.0	234.091
2	R	2.5	222.881
2	R	3.0	213.896
2	R	4.0	196.026
2	R	5.0	199.634
2	R	6.0	196.037
2	R	8.0	213.352
2	R	10.0	200.088
2	R	12.0	196.035
2	R	24.0	160.338
2	R	48.0	110.28
2	R	72.0	85.241
3	R	0.0	0.0
3	R	0.5	49.849
3	R	1.0	77.367
3	R	1.5	105.345
3	R	2.0	100.943
3	R	2.5	72.746
3	R	3.0	69.985
3	R	4.0	93.565
3	R	5.0	91.981
3	R	6.0	82.71
3	R	8.0	84.205

3	R	10.0	85.342
3	R	12.0	76.027
3	R	24.0	81.259
3	R	48.0	70.107
3	R	72.0	67.901
4	R	0.0	0.0
4	R	0.5	52.421
4	R	1.0	208.542
4	R	1.5	188.923
4	R	2.0	165.177
4	R	2.5	146.996
4	R	3.0	152.701
4	R	4.0	154.345
4	R	5.0	128.398
4	R	6.0	149.807
4	R	8.0	151.066
4	R	10.0	136.819
4	R	12.0	132.257
4	R	24.0	141.247
4	R	48.0	129.138
4	R	72.0	97.625
5	T	0.0	0.0
5	T	0.5	0.0
5	T	1.0	9.545
5	T	1.5	153.964
5	T	2.0	152.34
5	T	2.5	151.452
5	T	3.0	161.312
5	T	4.0	169.334
5	T	5.0	162.907
5	T	6.0	166.651
5	T	8.0	168.668
5	T	10.0	155.103
5	T	12.0	154.066
5	T	24.0	162.974
5	T	48.0	109.814
5	T	72.0	110.778
6	T	0.0	0.0
6	T	0.5	57.882
6	T	1.0	100.498
6	T	1.5	138.651
6	T	2.0	147.287
6	T	2.5	154.648
6	T	3.0	122.316
6	T	4.0	132.857
6	T	5.0	126.067
6	T	6.0	140.466
6	T	8.0	115.542
6	T	10.0	102.16

6	T	12.0	113.751
6	T	24.0	101.049
6	T	48.0	92.55
6	T	72.0	69.501
7	R	0.0	0.0
7	R	0.5	19.95
7	R	1.0	128.405
7	R	1.5	136.807
7	R	2.0	113.109
7	R	2.5	153.254
7	R	3.0	123.606
7	R	4.0	142.655
7	R	5.0	112.347
7	R	6.0	139.919
7	R	8.0	105.513
7	R	10.0	134.408
7	R	12.0	123.37
7	R	24.0	110.511
7	R	48.0	90.291
7	R	72.0	58.051
8	R	0.0	0.0
8	R	0.5	136.91
8	R	1.0	126.646
8	R	1.5	118.5
8	R	2.0	134.926
8	R	2.5	113.213
8	R	3.0	130.896
8	R	4.0	138.327
8	R	5.0	22.724
8	R	6.0	53.774
8	R	8.0	55.107
8	R	10.0	102.871
8	R	12.0	134.133
8	R	24.0	108.021
8	R	48.0	98.466
8	R	72.0	74.437
9	T	0.0	0.0
9	T	0.5	113.362
9	T	1.0	128.273
9	T	1.5	125.395
9	T	2.0	146.933
9	T	2.5	140.559
9	T	3.0	167.347
9	T	4.0	157.504
9	T	5.0	141.35
9	T	6.0	140.282
9	T	8.0	105.438
9	T	10.0	164.843
9	T	12.0	135.58

9	T	24.0	117.125
9	T	48.0	109.745
9	T	72.0	93.44
10	R	0.0	0.0
10	R	0.5	13.634
10	R	1.0	62.561
10	R	1.5	112.655
10	R	2.0	125.482
10	R	2.5	116.255
10	R	3.0	112.674
10	R	4.0	116.986
10	R	5.0	119.81
10	R	6.0	107.557
10	R	8.0	120.479
10	R	10.0	124.171
10	R	12.0	106.476
10	R	24.0	116.508
10	R	48.0	45.204
10	R	72.0	42.191

12.0.2 Testing urine PK dataset.

subj	conc	st	et	vol
1	1	0	1	1
1	2	1	2	2
1	2	2	6	3
1	1	6	12	3
1	1	12	18	2

12.0.3 Testing PD dataset.

subj	time	obs
1	0.0	0
1	1.0	1
1	2.0	4
1	2.5	7
1	3.0	5
1	3.3	4
1	3.6	3
1	4.0	2
1	5.0	8
1	6.0	1
1	7.0	2
1	8.0	1
1	9.0	1

13 Appendix 2

13.0.1 Reference output.

Avialible at https://github.com/PharmCat/MetidaNCA.jl/tree/main/docs/src/pdf.

See Appendix 2.1.pdf,

Appendix 2.2.pdf,

Appendix 2.3.pdf,

Appendix 2.4.pdf,

Appendix 2.5.pdf.